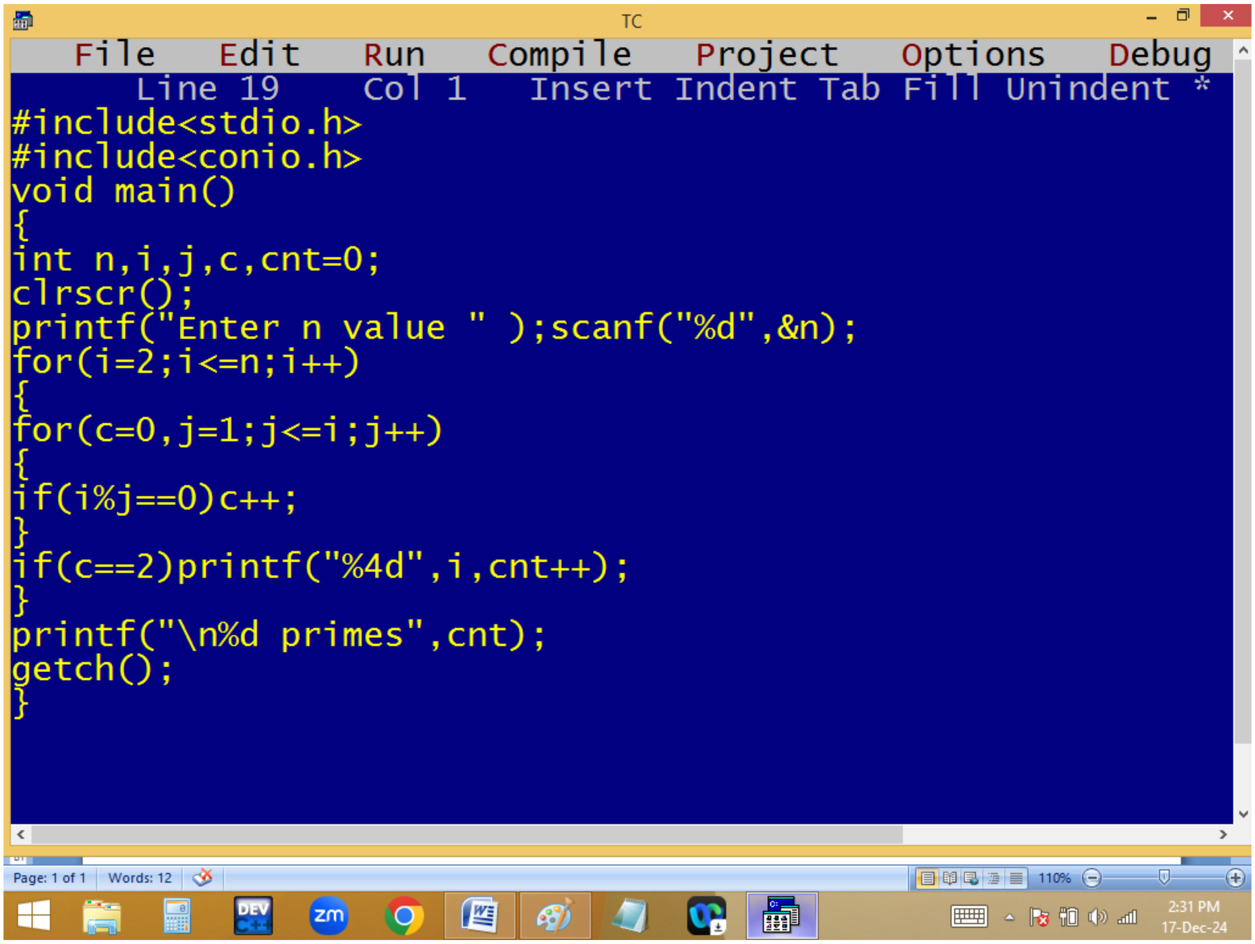
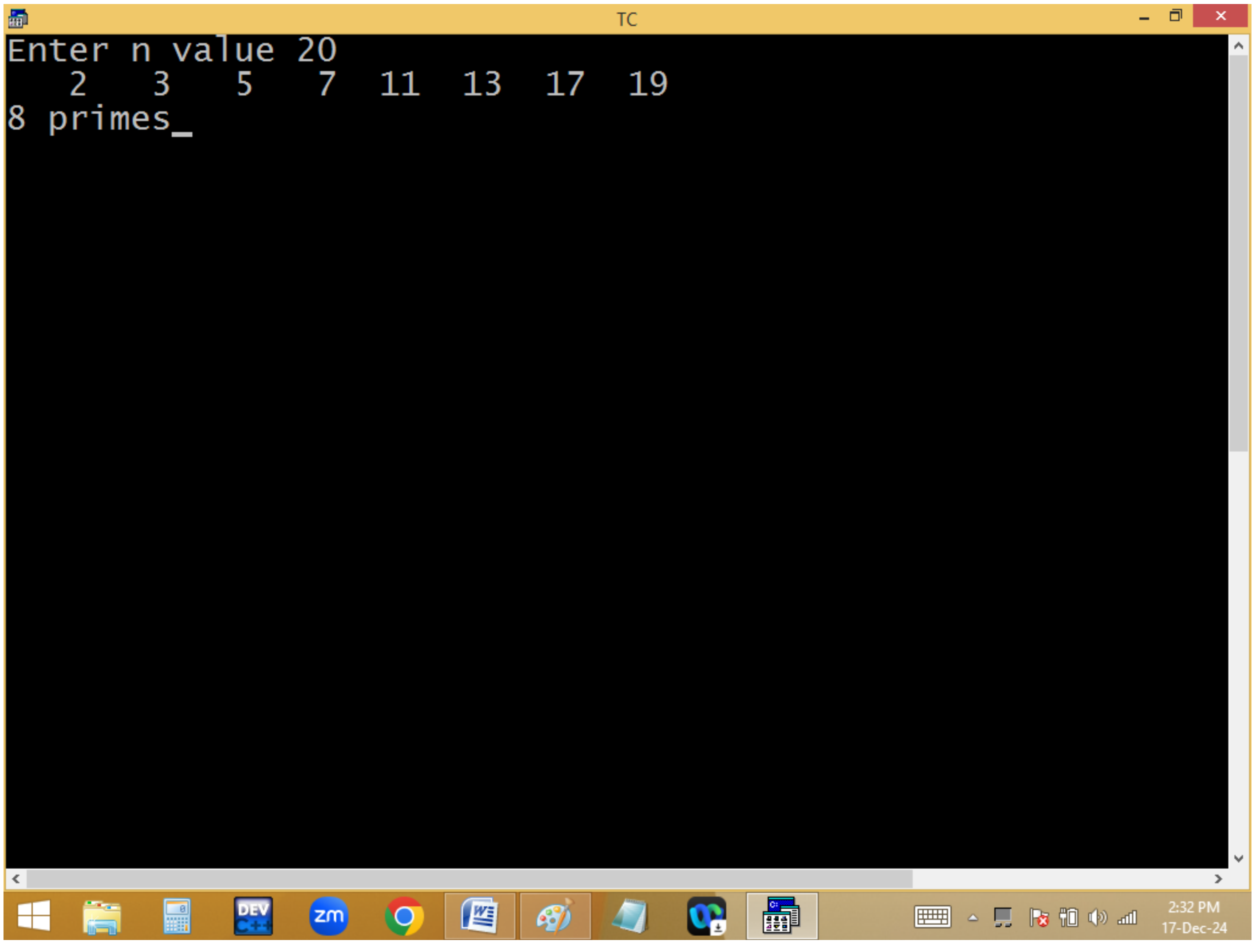
Finding 1..n primes and count.



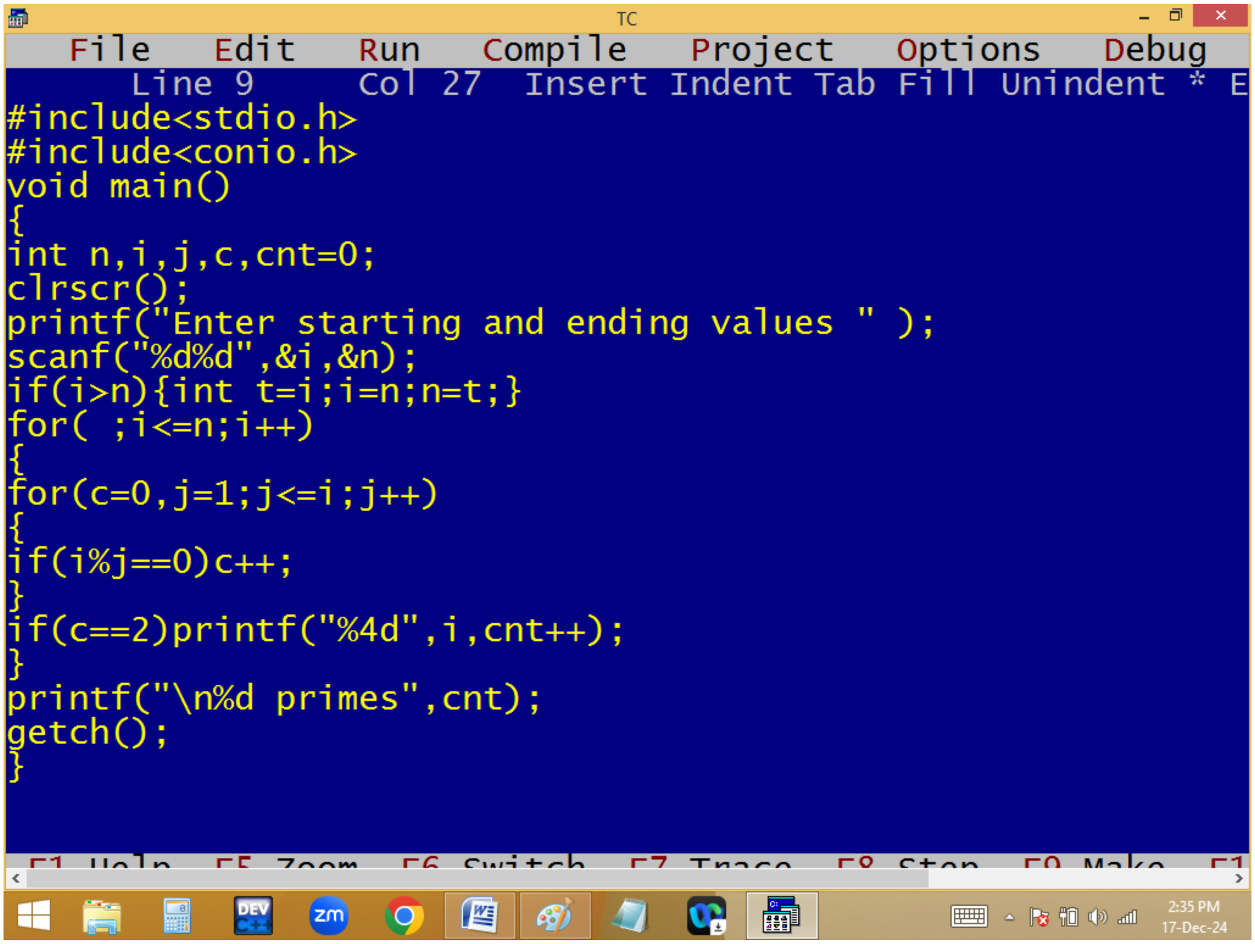

```c
#include<stdio.h>
#include<conio.h>
void main()
{
int n,i,j,c,cnt=0;
clrscr();
printf("Enter n value " );scanf("%d",&n);
for(i=2;i<=n;i++)
{
for(c=0,j=1;j<=i;j++)
{
if(i%j==0)c++;
}
if(c==2)printf("%4d",i,cnt++);
}
printf("\n%d primes",cnt);
getch();
}
```



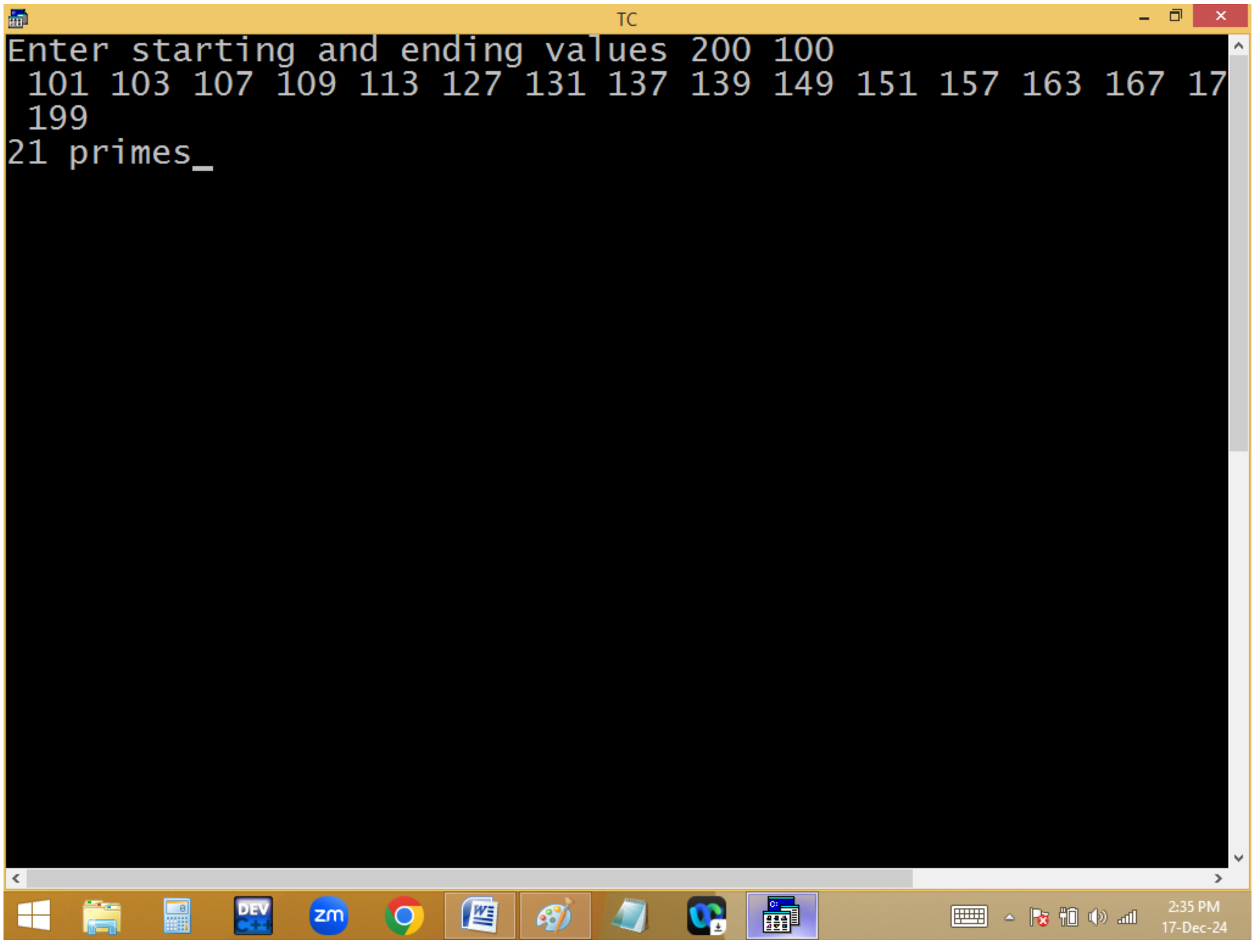

```
cnt=0;
for(i=2; i<=n;i++)
{
for(c=0, j=1;j<=i;j++)
{
if(i%j==0)c++;
}
if( c==2 ) p(i,cnt++);
}
p(cnt);
```

n	i	j	c	cnt
10	2 %	1=0	0 1 2	0
	2 %	2=0		1
	3 %	1=0		
	3 %	3=0	0 1 2	2
	4 %	1=0	0 1 2 3	
		2=0		
		4=0		

Finding n to n primes and count




```c
#include<stdio.h>
#include<conio.h>
void main()
{
int n,i,j,c,cnt=0;
clrscr();
printf("Enter starting and ending values " );
scanf("%d%d",&i,&n);
if(i>n){int t=i;i=n;n=t;}
for( ;i<=n;i++)
{
for(c=0,j=1;j<=i;j++)
{
if(i%j==0)c++;
}
if(c==2)printf("%4d",i,cnt++);
}
printf("\n%d primes",cnt);
getch();
}
```

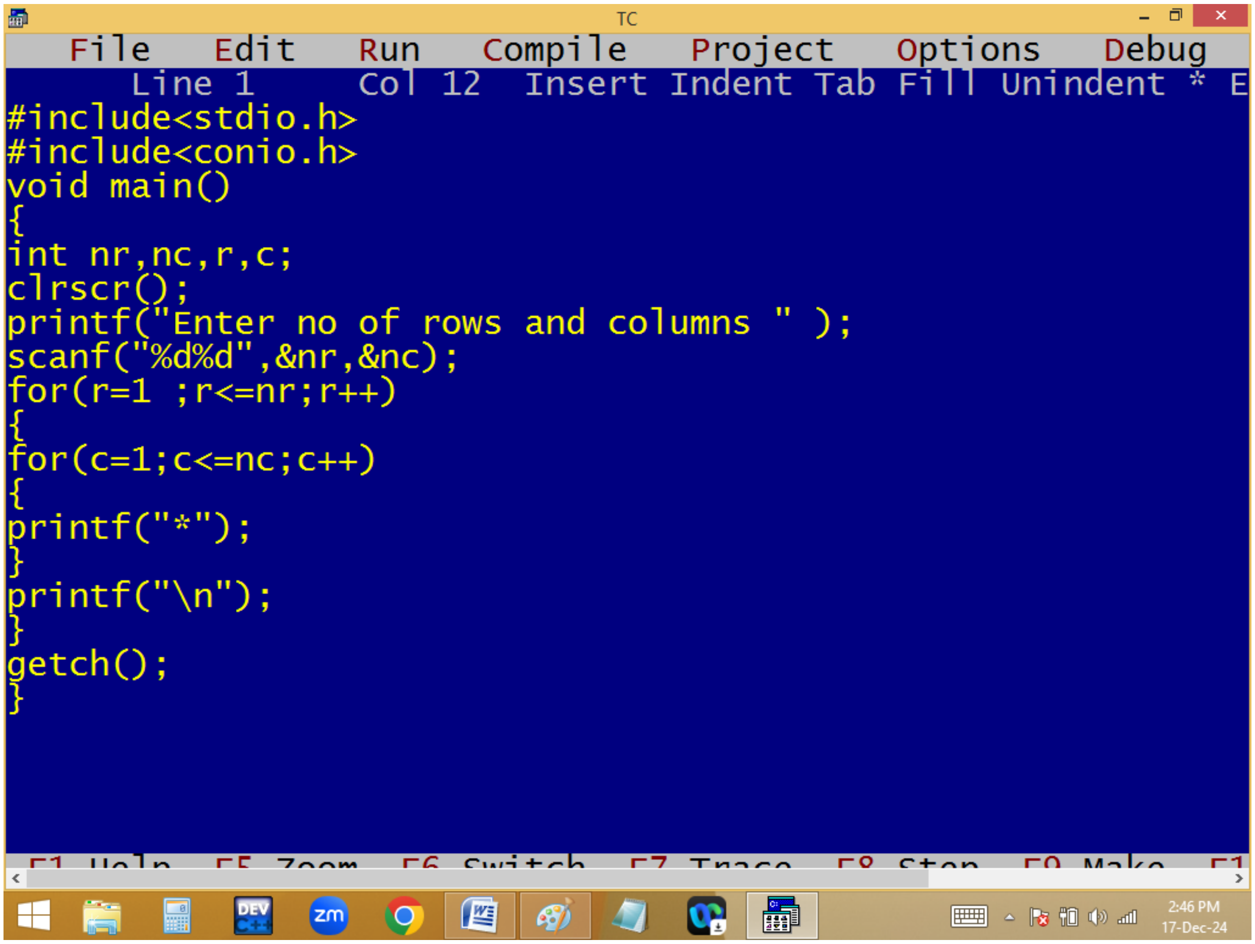

TC
Enter starting and ending values 200 100
101 103 107 109 113 127 131 137 139 149 151 157 163 167 17
199
21 primes_
2:35 PM
17-Dec-24

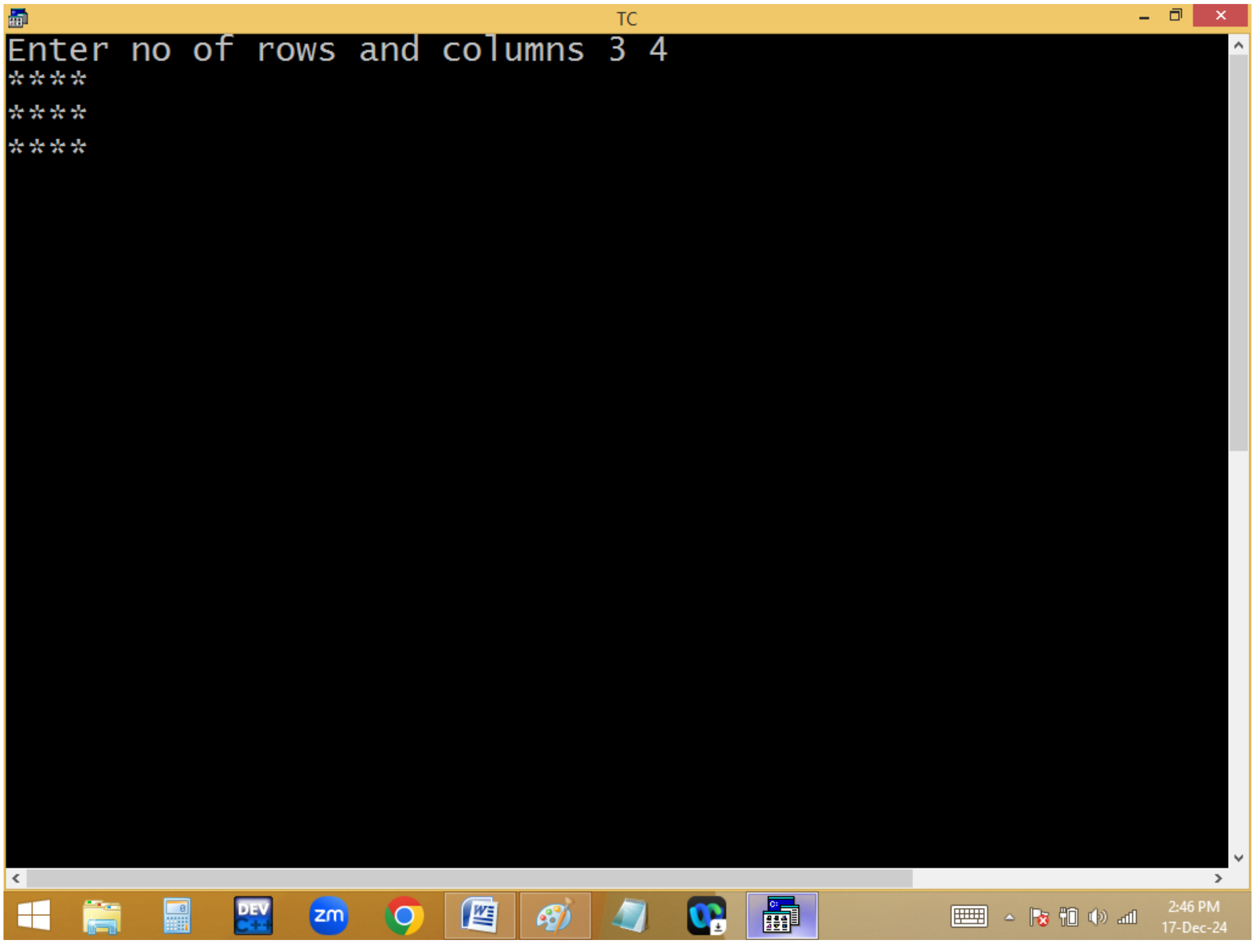
Patterns:

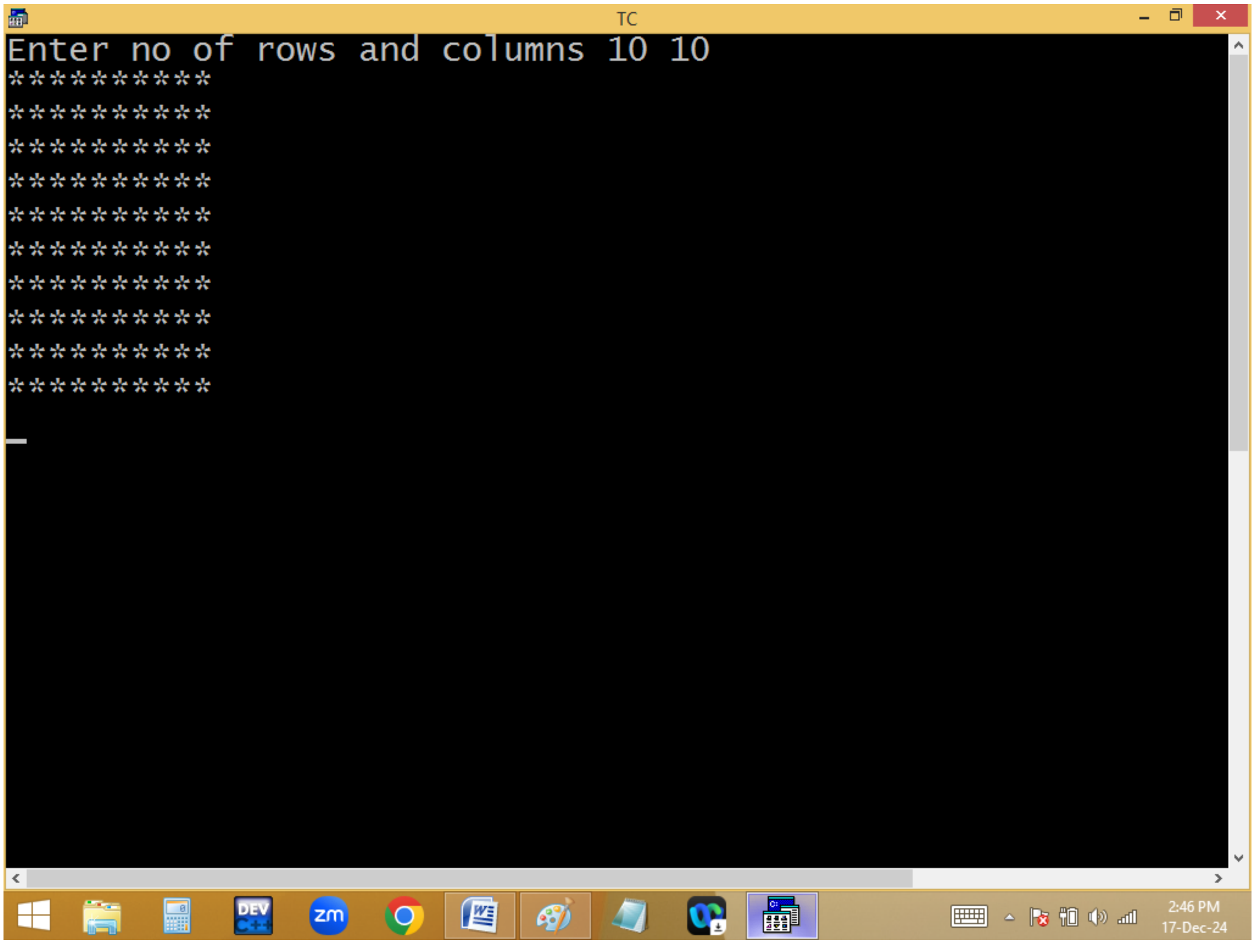
* * * *

* * * *

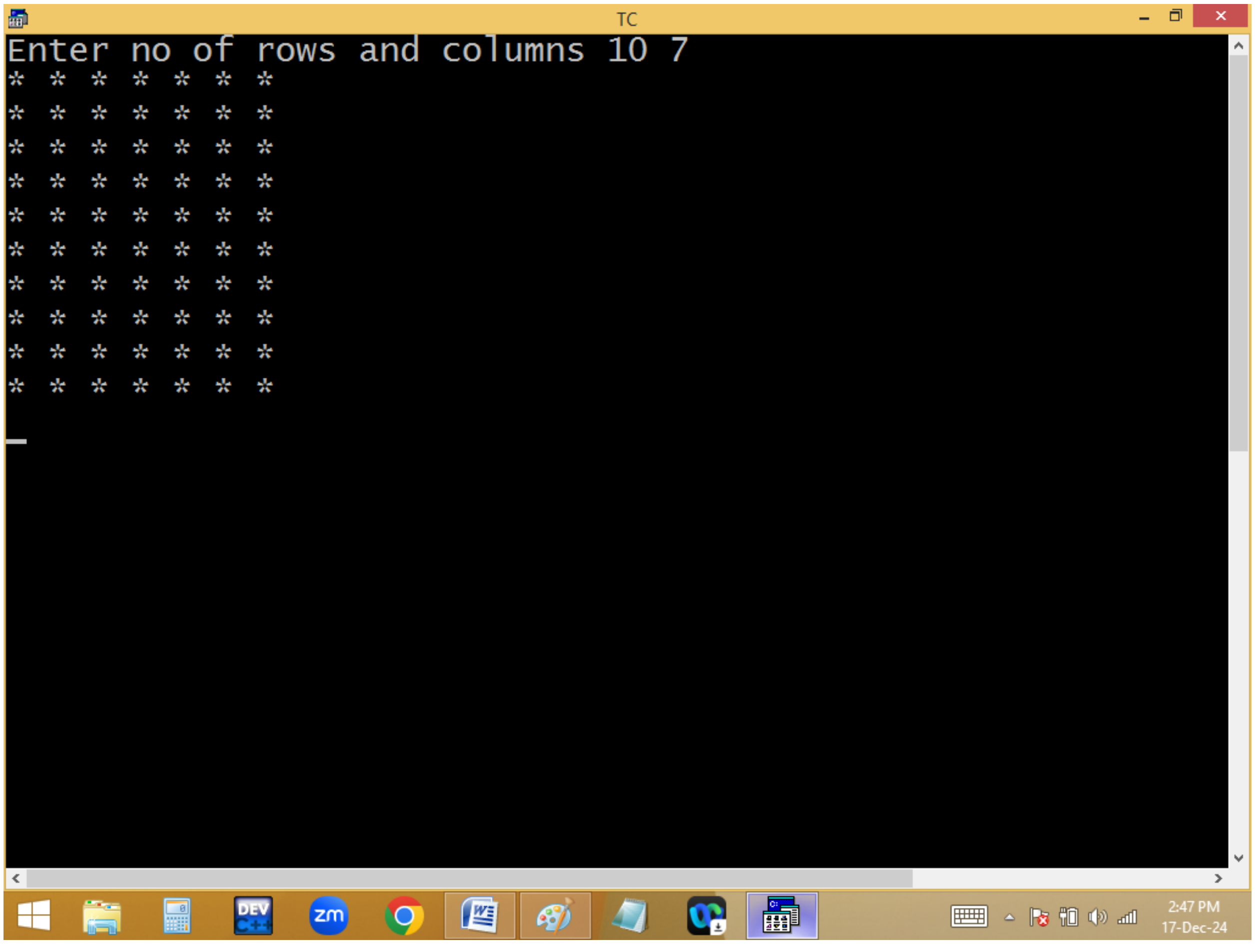
* * * *

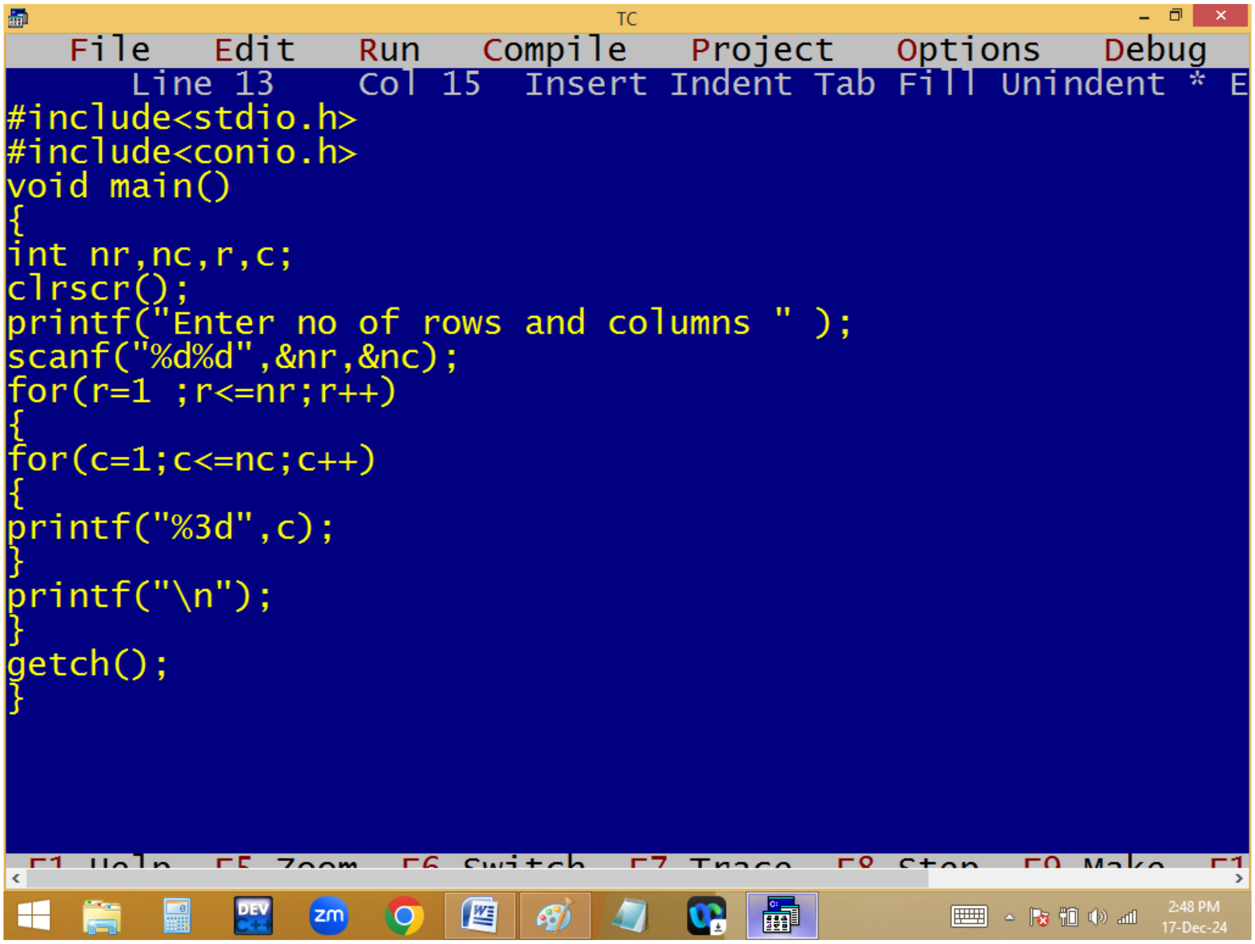

TC
File Edit Run Compile Project Options Debug
Line 1 Col 12 Insert Indent Tab Fill Unindent * E
#include<stdio.h>
#include<conio.h>
void main()
{
int nr,nc,r,c;
clrscr();
printf("Enter no of rows and columns ");
scanf("%d%d",&nr,&nc);
for(r=1 ;r<=nr;r++)
{
for(c=1;c<=nc;c++)
{
printf("*");
}
printf("\n");
}
getch();
}
2:46 PM
17-Dec-24


TC
Enter no of rows and columns 3 4
****
****
****
2:46 PM
17-Dec-24

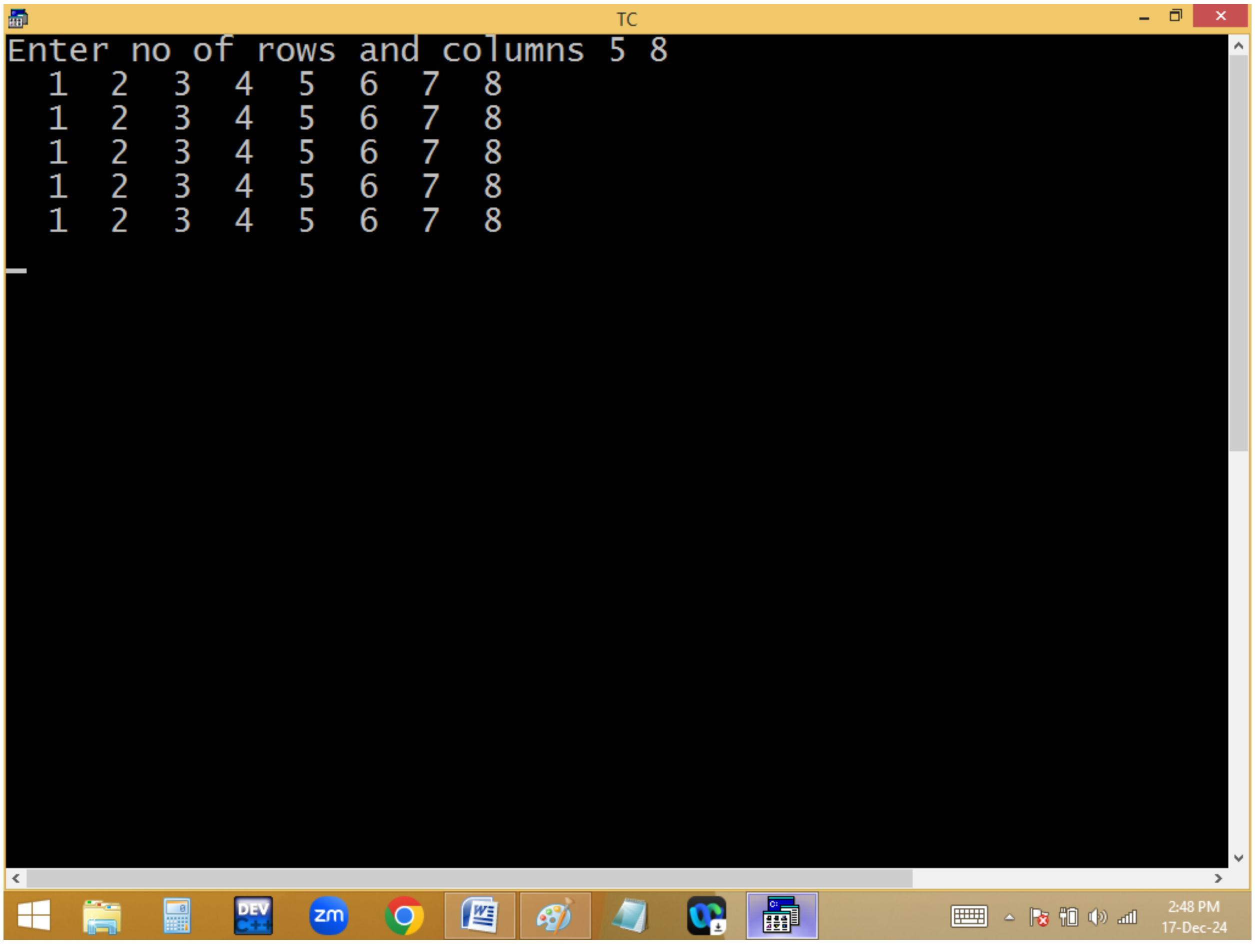

TC
Enter no of rows and columns 10 10
****
****
****
****
****
****
****
****
****
****
2:46 PM
17-Dec-24

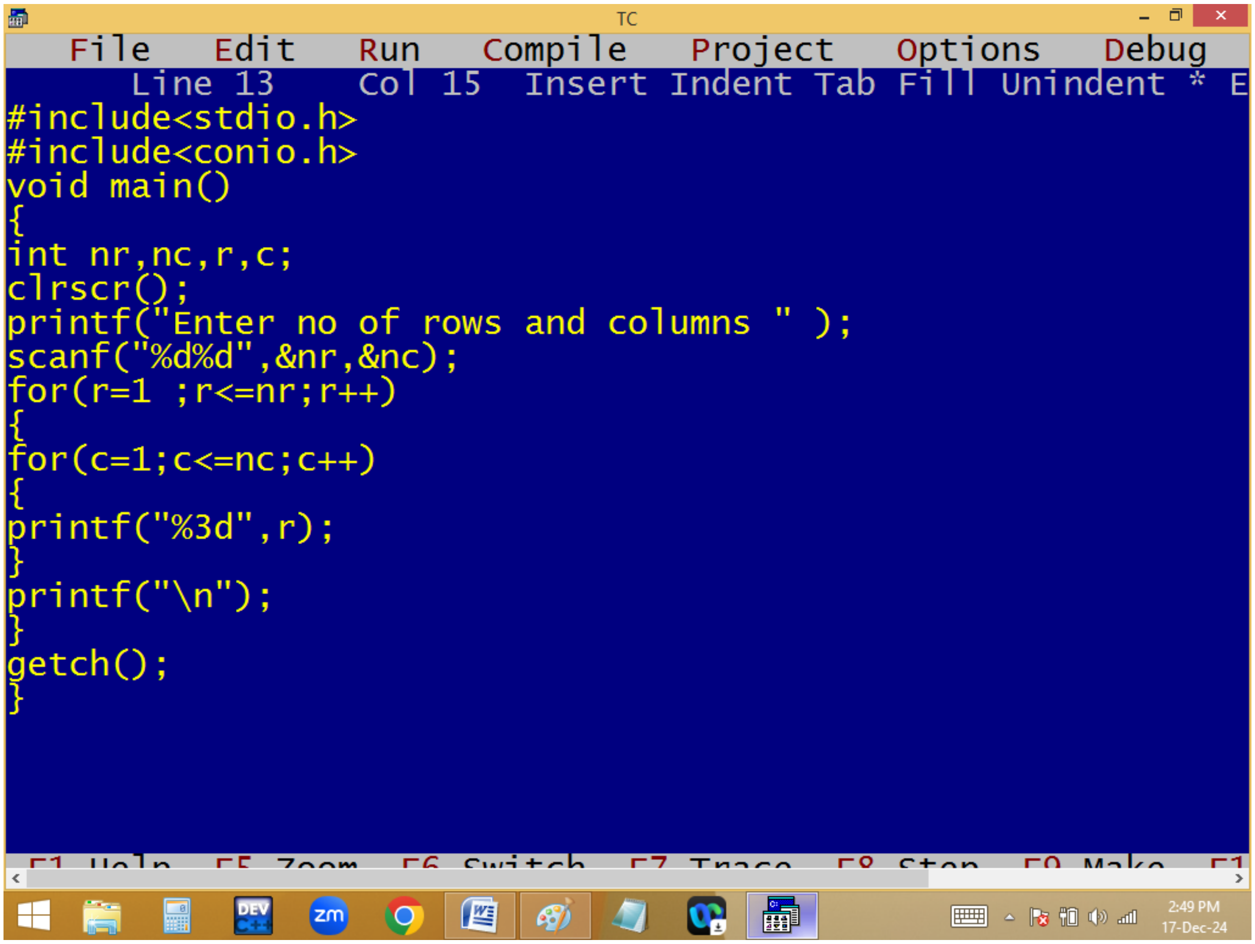
```
TC
    File    Edit    Run    Compile    Project    Options    Debug
        Line 13     Col 11   Insert Indent Tab Fill Unindent * E
#include<stdio.h>
#include<conio.h>
void main()
{
int nr,nc,r,c;
clrscr();
printf("Enter no of rows and columns " );
scanf("%d%d",&nr,&nc);
for(r=1 ;r<=nr;r++)
{
for(c=1;c<=nc;c++)
{
printf("* ");
}
printf("\n");
}
getch();
}
```

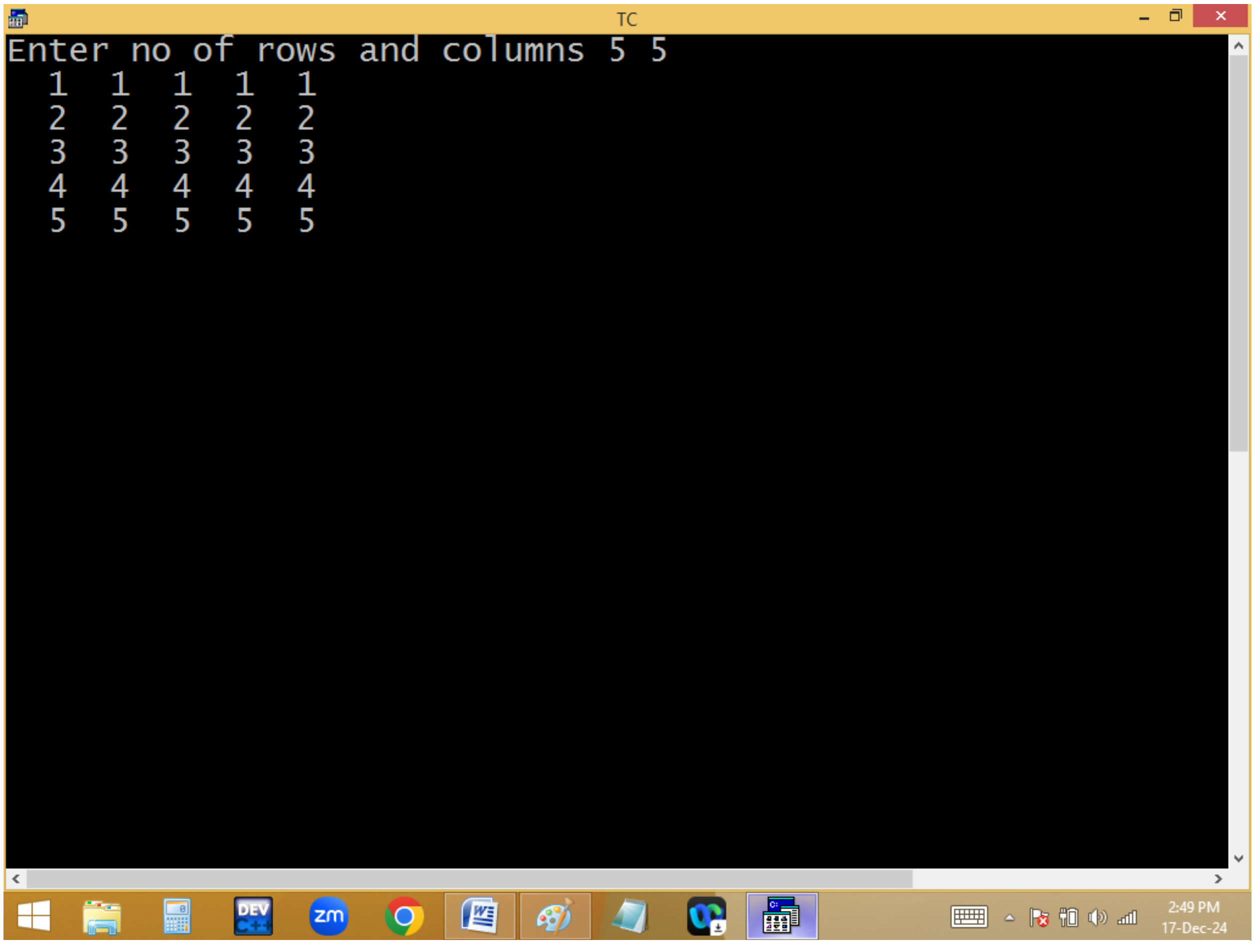

TC
Enter no of rows and columns 10 7
* * * * * * *
* * * * * * *
* * * * * * *
* * * * * * *
* * * * * * *
* * * * * * *
* * * * * * *
* * * * * * *
* * * * * * *
* * * * * * *
2:47 PM
17-Dec-24



```c
#include<stdio.h>
#include<conio.h>
void main()
{
int nr,nc,r,c;
clrscr();
printf("Enter no of rows and columns " );
scanf("%d%d",&nr,&nc);
for(r=1 ;r<=nr;r++)
{
for(c=1;c<=nc;c++)
{
printf("%3d",c);
}
printf("\n");
}
getch();
}
```

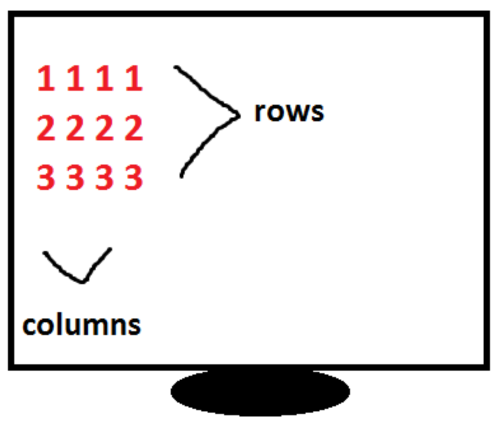

TC
Enter no of rows and columns 5 8
1 2 3 4 5 6 7 8
1 2 3 4 5 6 7 8
1 2 3 4 5 6 7 8
1 2 3 4 5 6 7 8
1 2 3 4 5 6 7 8
2:48 PM
17-Dec-24

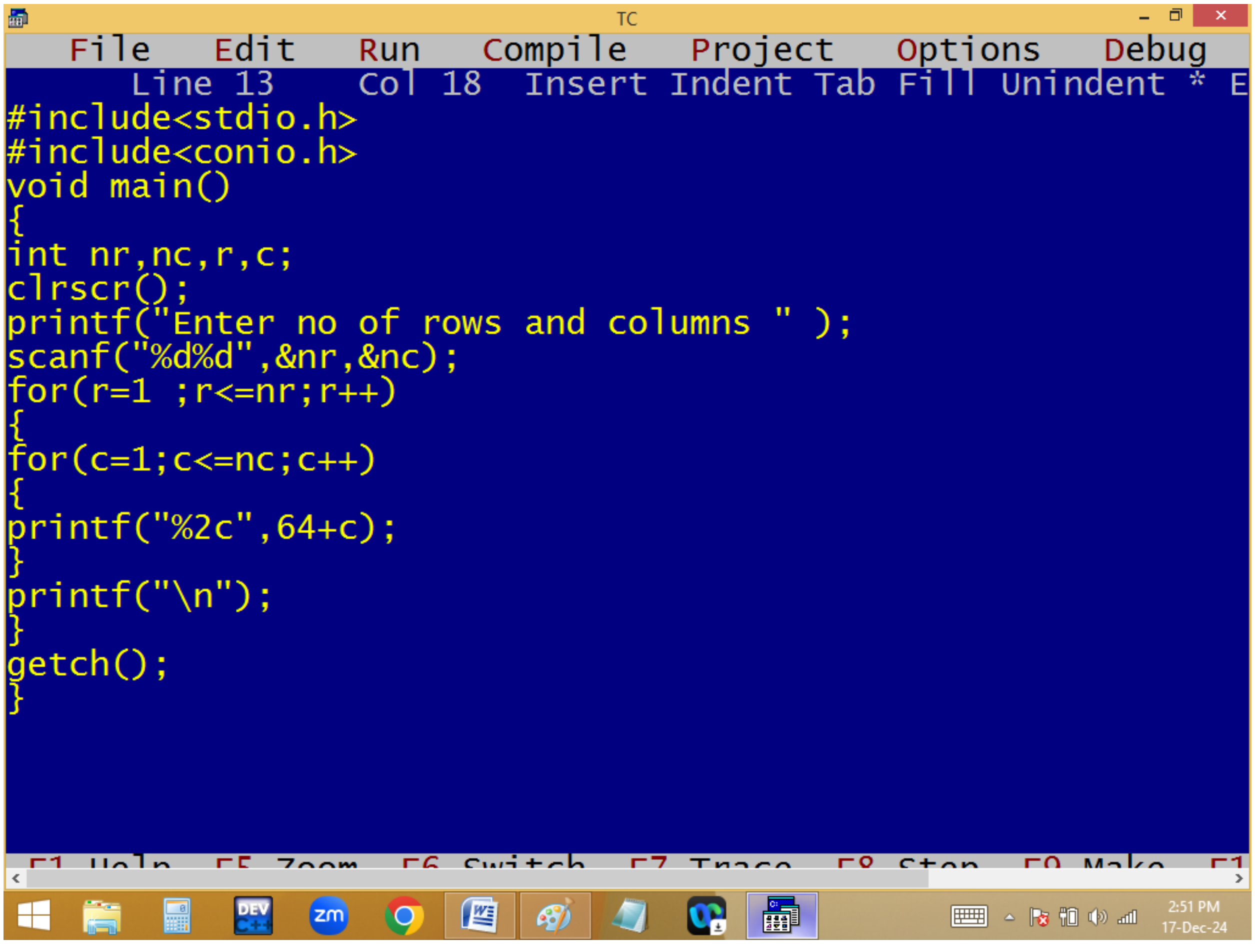

TC
File Edit Run Compile Project Options Debug
Line 13 Col 15 Insert Indent Tab Fill Unindent * E
#include<stdio.h>
#include<conio.h>
void main()
{
int nr,nc,r,c;
clrscr();
printf("Enter no of rows and columns ");
scanf("%d%d",&nr,&nc);
for(r=1 ;r<=nr;r++)
{
for(c=1;c<=nc;c++)
{
printf("%3d",r);
}
printf("\n");
}
getch();
}
2:49 PM
17-Dec-24



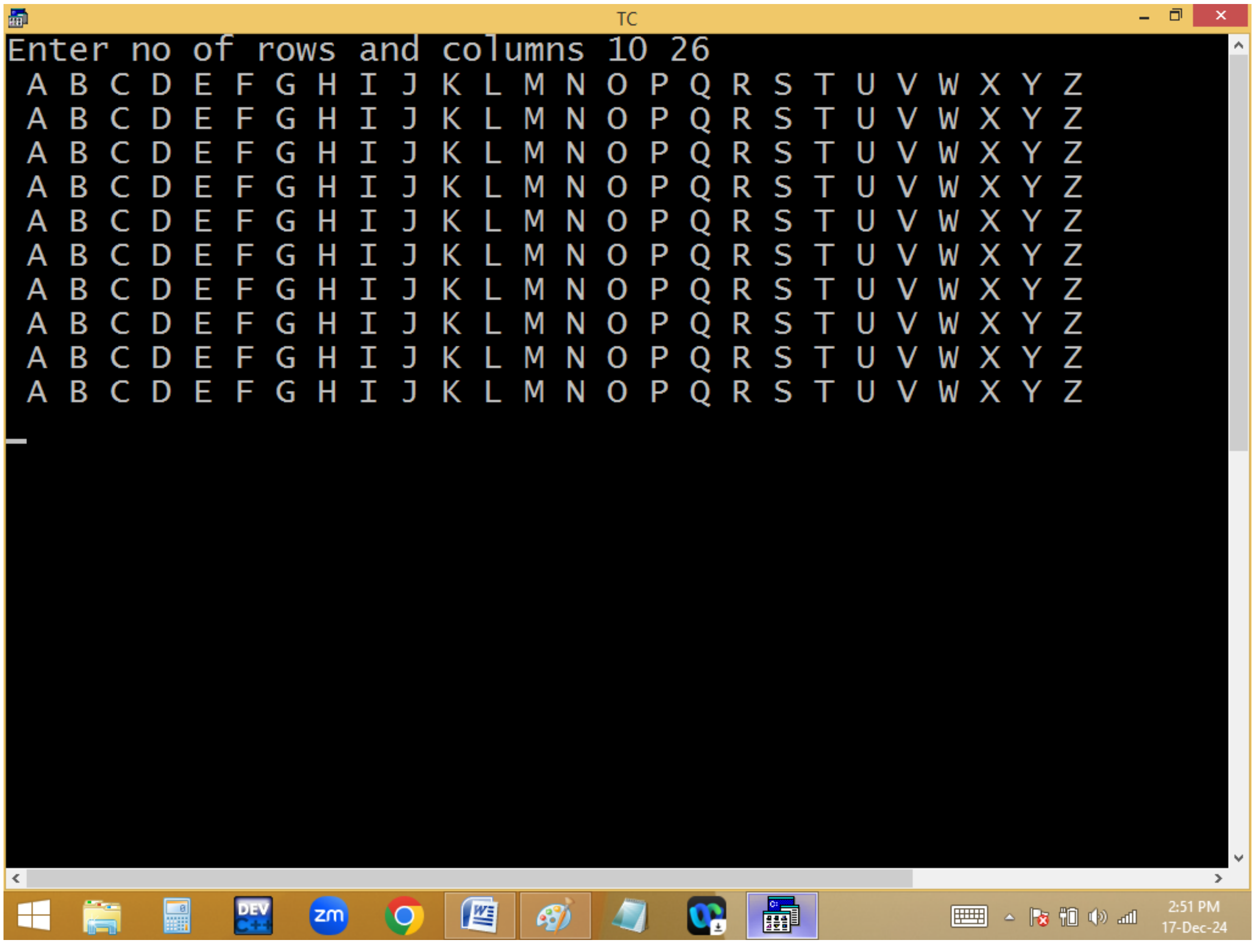

```
for( r=1; r<=3; r++)
{
for( c=1;c<=4;c++)
{
 p( r );
}

 p("\n");
}
```

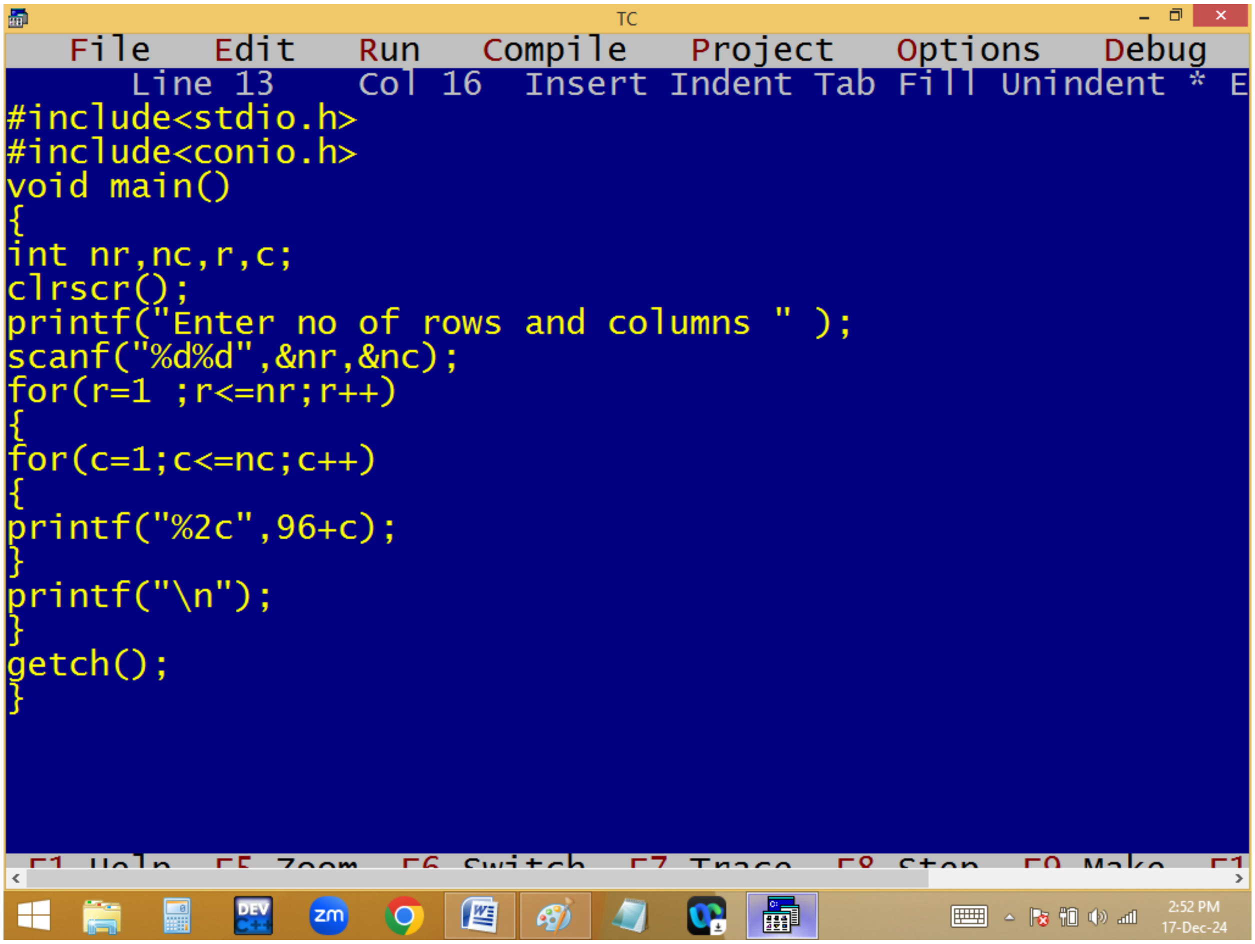
r	c				
1	1	2	3	4	5
	*	*	*	*	
2	1	2	3	4	5
	*	*	*	*	
3	1	2	3	4	5
	*	*	*	*	
~~4~~					



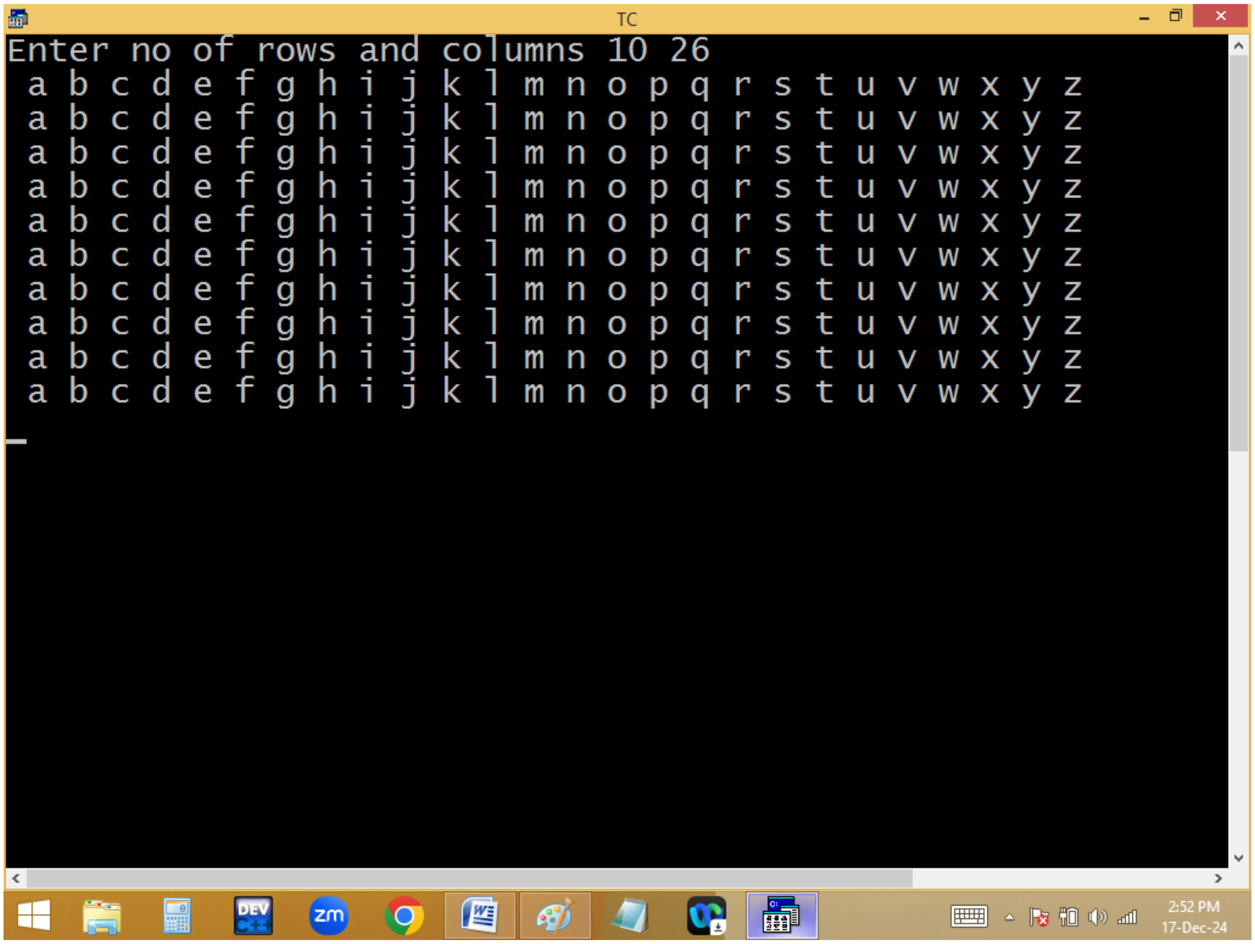


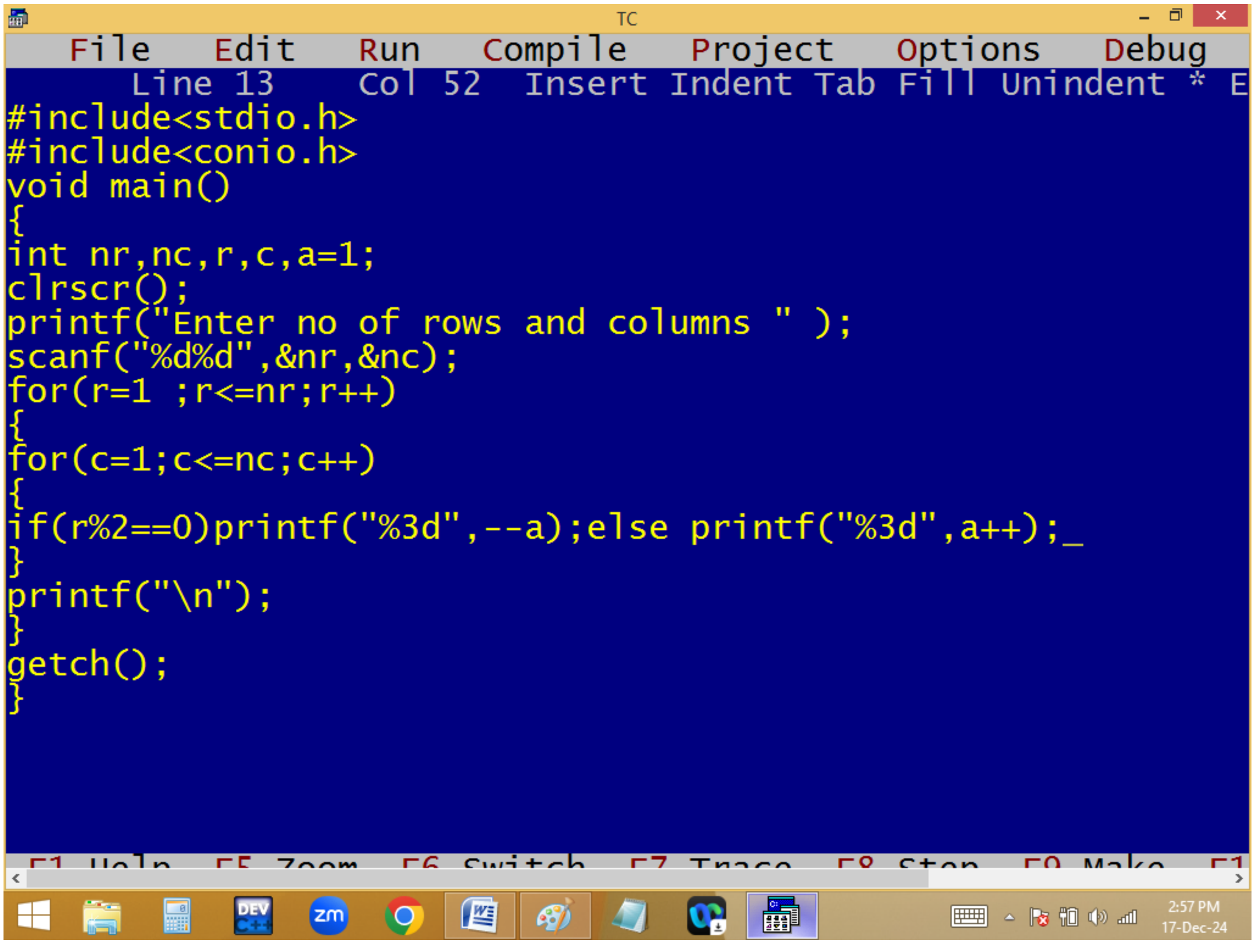

```
#include<stdio.h>
#include<conio.h>
void main()
{
int nr,nc,r,c;
clrscr();
printf("Enter no of rows and columns " );
scanf("%d%d",&nr,&nc);
for(r=1 ;r<=nr;r++)
{
for(c=1;c<=nc;c++)
{
printf("%2c",64+c);
}
printf("\n");
}
getch();
}
```

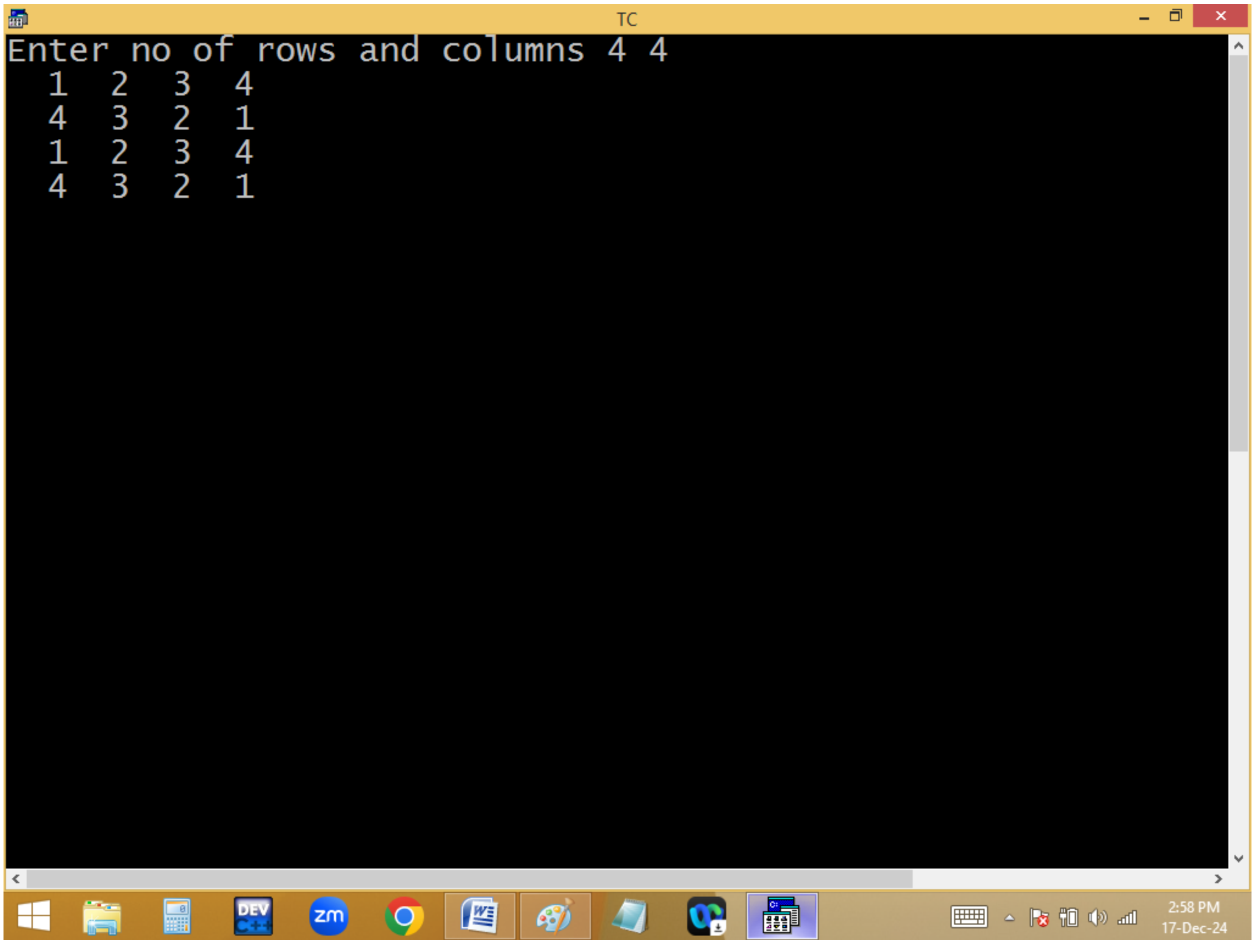

TC
Enter no of rows and columns 10 26
A B C D E F G H I J K L M N O P Q R S T U V W X Y Z
A B C D E F G H I J K L M N O P Q R S T U V W X Y Z
A B C D E F G H I J K L M N O P Q R S T U V W X Y Z
A B C D E F G H I J K L M N O P Q R S T U V W X Y Z
A B C D E F G H I J K L M N O P Q R S T U V W X Y Z
A B C D E F G H I J K L M N O P Q R S T U V W X Y Z
A B C D E F G H I J K L M N O P Q R S T U V W X Y Z
A B C D E F G H I J K L M N O P Q R S T U V W X Y Z
A B C D E F G H I J K L M N O P Q R S T U V W X Y Z
A B C D E F G H I J K L M N O P Q R S T U V W X Y Z
2:51 PM
17-Dec-24




```c
#include<stdio.h>
#include<conio.h>
void main()
{
int nr,nc,r,c;
clrscr();
printf("Enter no of rows and columns " );
scanf("%d%d",&nr,&nc);
for(r=1 ;r<=nr;r++)
{
for(c=1;c<=nc;c++)
{
printf("%2c",96+c);
}
printf("\n");
}
getch();
}
```

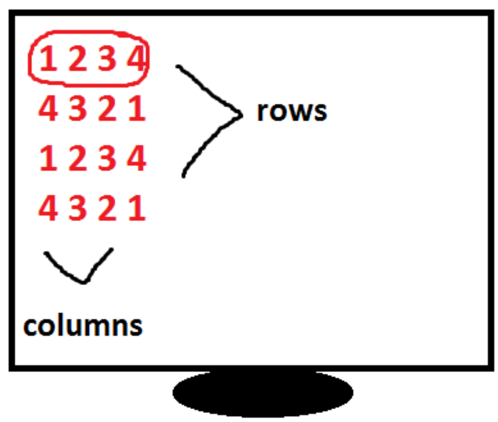

TC
Enter no of rows and columns 10 26
a b c d e f g h i j k l m n o p q r s t u v w x y z
a b c d e f g h i j k l m n o p q r s t u v w x y z
a b c d e f g h i j k l m n o p q r s t u v w x y z
a b c d e f g h i j k l m n o p q r s t u v w x y z
a b c d e f g h i j k l m n o p q r s t u v w x y z
a b c d e f g h i j k l m n o p q r s t u v w x y z
a b c d e f g h i j k l m n o p q r s t u v w x y z
a b c d e f g h i j k l m n o p q r s t u v w x y z
a b c d e f g h i j k l m n o p q r s t u v w x y z
a b c d e f g h i j k l m n o p q r s t u v w x y z
2:52 PM
17-Dec-24


TC
File Edit Run Compile Project Options Debug
Line 13 Col 52 Insert Indent Tab Fill Unindent * E
#include<stdio.h>
#include<conio.h>
void main()
{
int nr,nc,r,c,a=1;
clrscr();
printf("Enter no of rows and columns ");
scanf("%d%d",&nr,&nc);
for(r=1 ;r<=nr;r++)
{
for(c=1;c<=nc;c++)
{
if(r%2==0)printf("%3d",--a);else printf("%3d",a++);_
}
printf("\n");
}
getch();
}
2:57 PM
17-Dec-24




```
int a=1;
for( r=1; r<=3; r++)
{
for( c=1;c<=4;c++)
{
p(r%2==0) p(--a);
else p(a++);
}
p("\n");
}
```

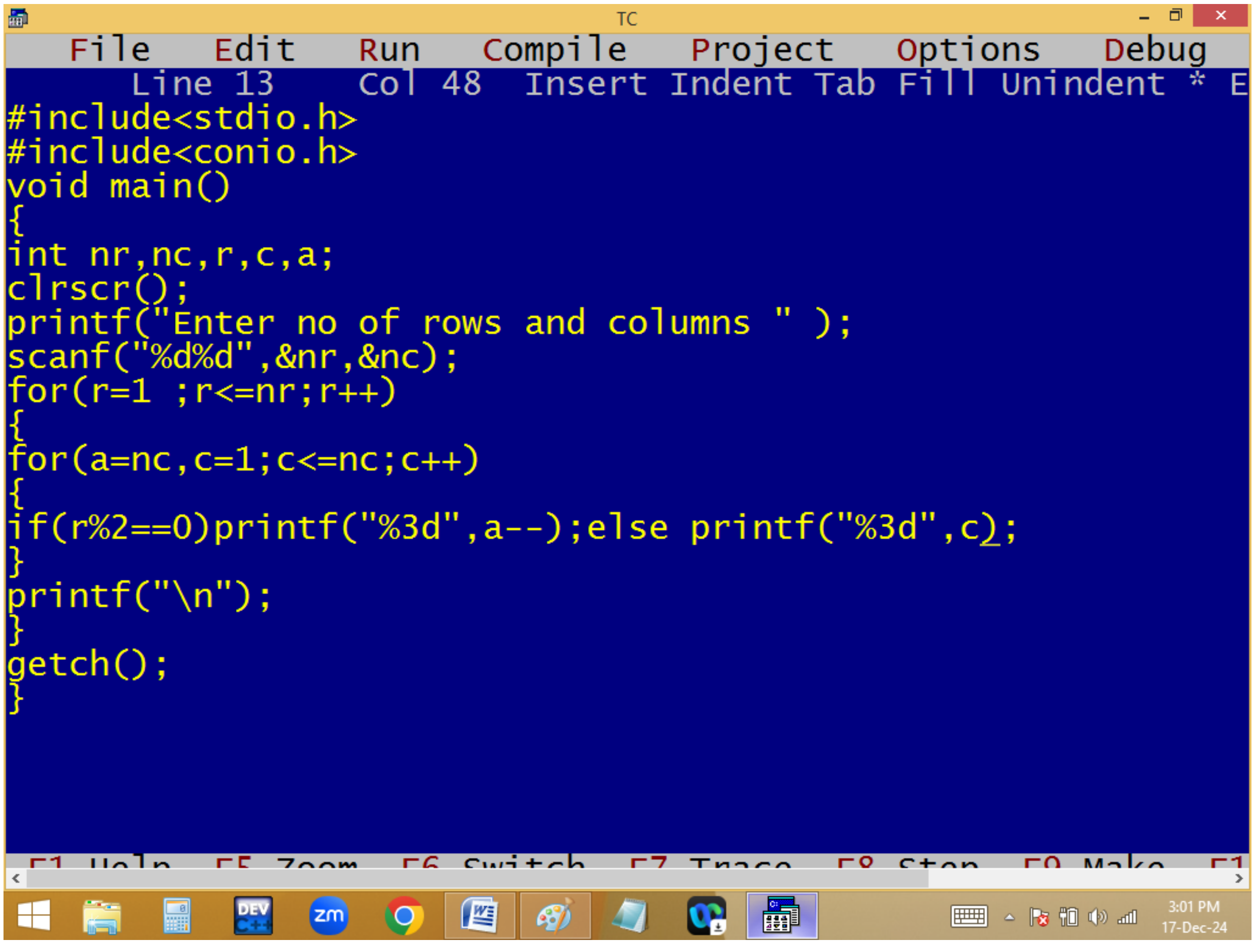
r	c
1	1 2 3 4 ~~5~~
	✗ ✗ ✗ ✗
2	1 2 3 4 ~~5~~
	✗ ✗ ✗ ✗
3	1 2 3 4 ~~5~~
	✗ ✗ ✗ ✗
~~4~~	

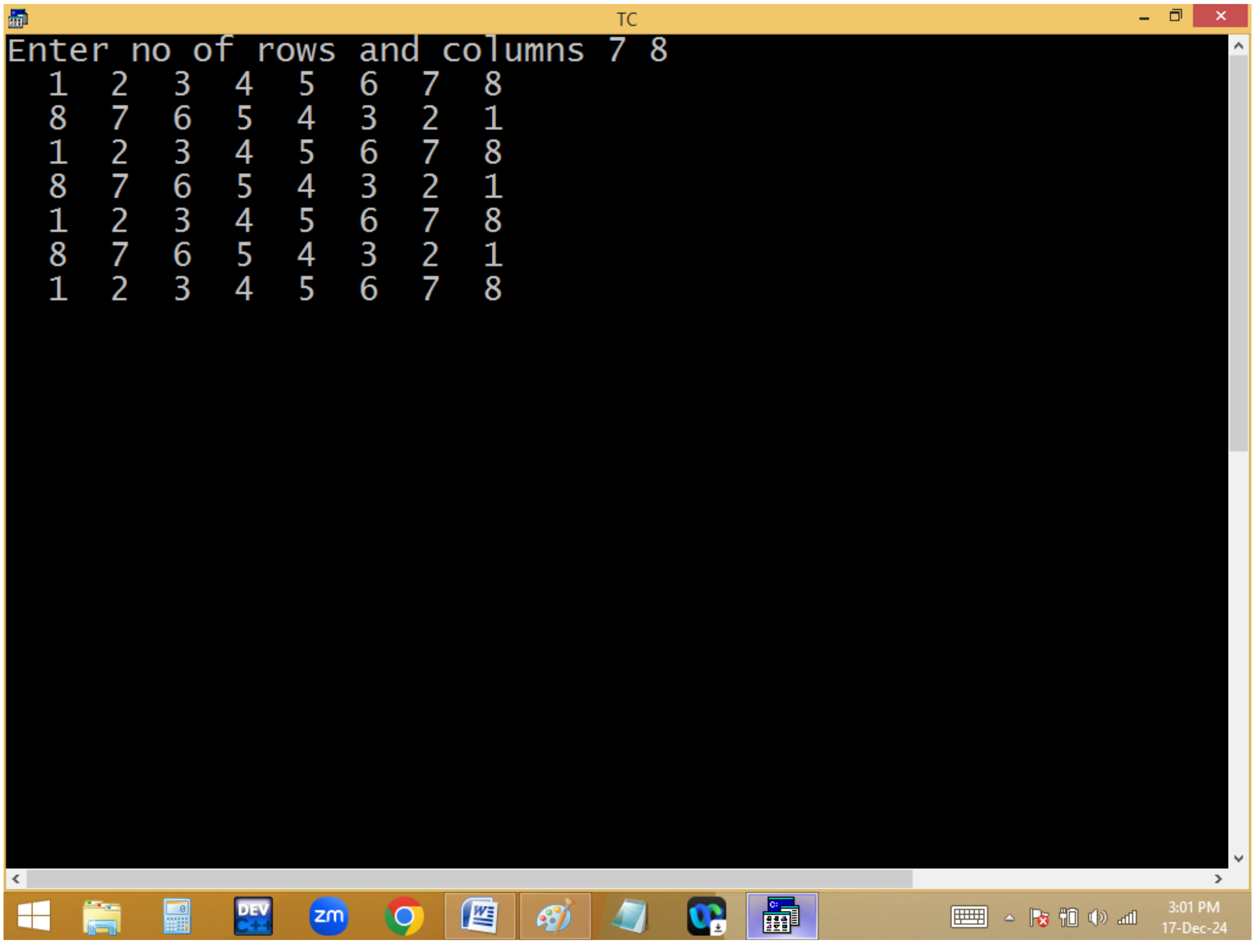


a

1 2 3 4 5

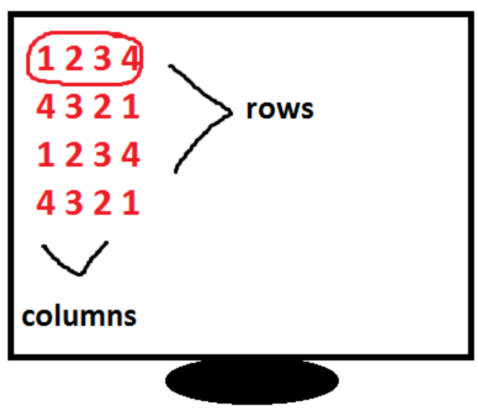
4 3 2 1


TC
File Edit Run Compile Project Options Debug
Line 13 Col 48 Insert Indent Tab Fill Unindent * E
#include<stdio.h>
#include<conio.h>
void main()
{
int nr,nc,r,c,a;
clrscr();
printf("Enter no of rows and columns ");
scanf("%d%d",&nr,&nc);
for(r=1 ;r<=nr;r++)
{
for(a=nc,c=1;c<=nc;c++)
{
if(r%2==0)printf("%3d",a--);else printf("%3d",c);
}
printf("\n");
}
getch();
}
3:01 PM
17-Dec-24




```
   int a   ;
for( r=1; r<=3; r++)
{ a=nc;
for( c=1;c<=4;c++)
{
p(r%2==0) p(a--);
else p( c  );
}
p("\n");
}
```

r c
1 1 2 3 4 5
x x x x
2 1 2 3 4 5
x x x x
3 1 2 3 4 5
x x x x
~~4~~



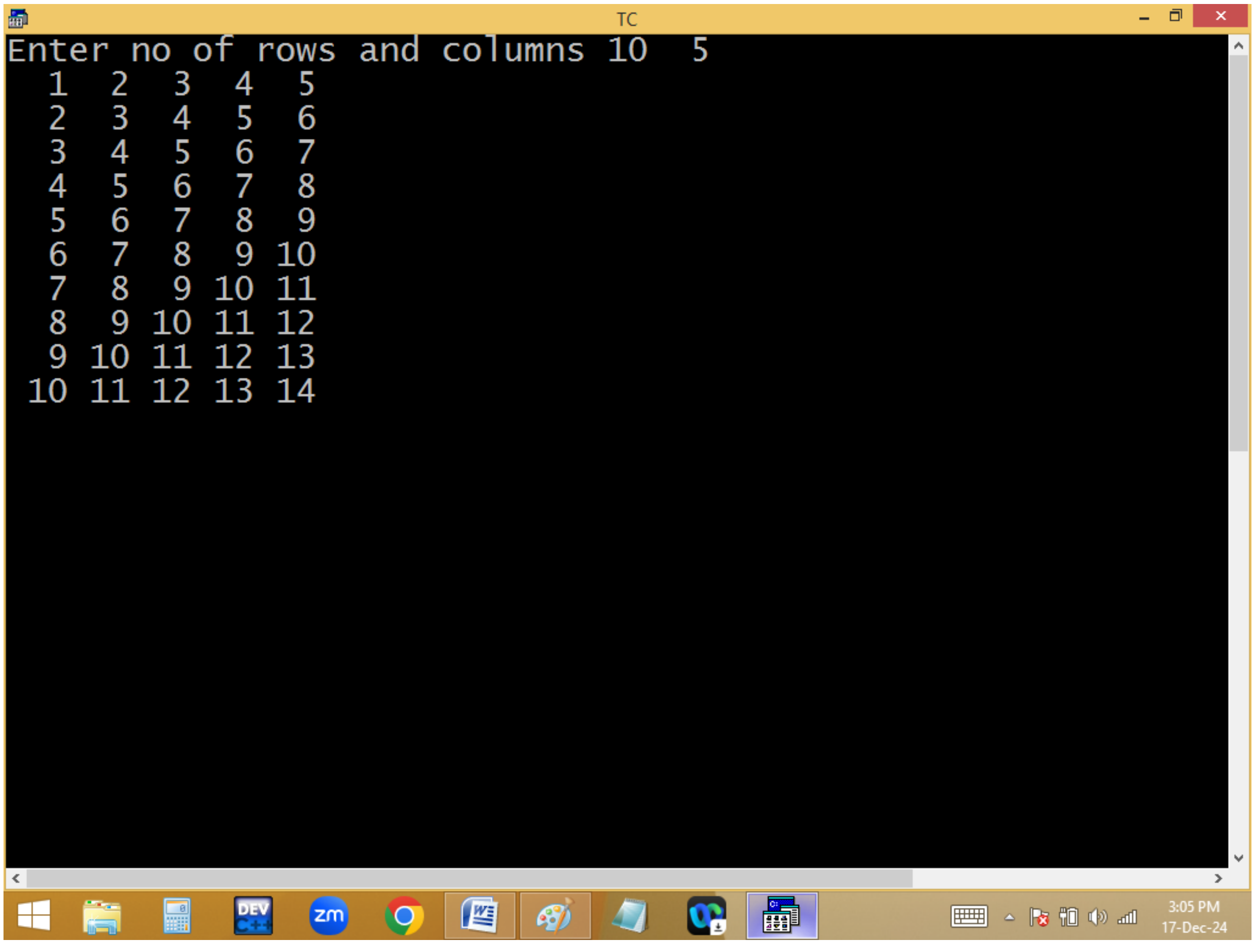
a

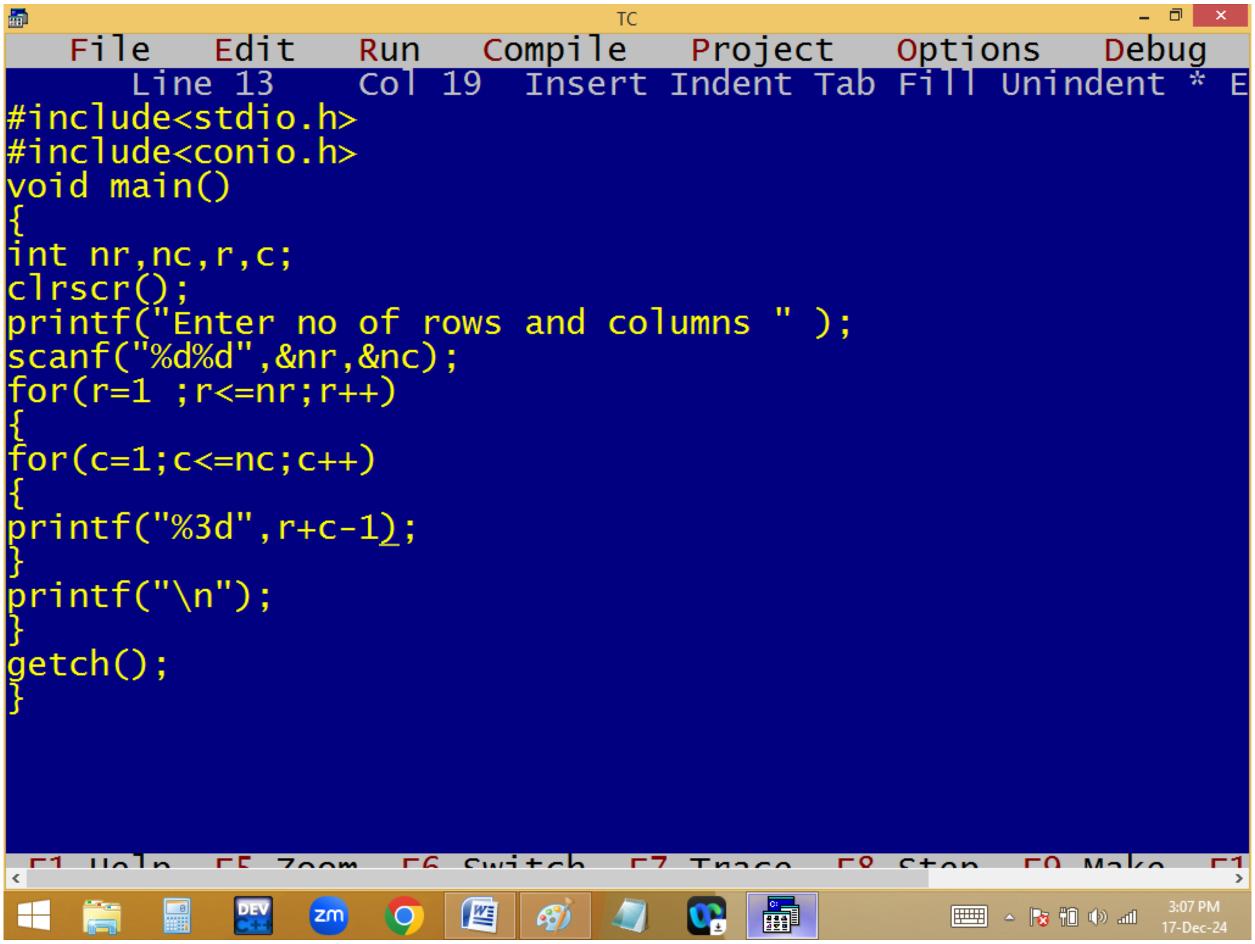
```
    File    Edit    Run    Compile    Project    Options    Debug
       Line 1      Col 18   Insert Indent Tab Fill Unindent * E
#include<stdio.h>
#include<conio.h>
void main()
{
int nr,nc,r,c,a;
clrscr();
printf("Enter no of rows and columns " );
scanf("%d%d",&nr,&nc);
for(r=1 ;r<=nr;r++)
{
for(a=r,c=1;c<=nc;c++)
{
printf("%3d",a++);
}
printf("\n");
}
getch();
}
```

TC

```
Enter no of rows and columns 5 9
   1   2   3   4   5   6   7   8   9
   2   3   4   5   6   7   8   9  10
   3   4   5   6   7   8   9  10  11
   4   5   6   7   8   9  10  11  12
   5   6   7   8   9  10  11  12  13
```

3:05 PM
17-Dec-24


TC
Enter no of rows and columns 10 5
1 2 3 4 5
2 3 4 5 6
3 4 5 6 7
4 5 6 7 8
5 6 7 8 9
6 7 8 9 10
7 8 9 10 11
8 9 10 11 12
9 10 11 12 13
10 11 12 13 14
3:05 PM
17-Dec-24

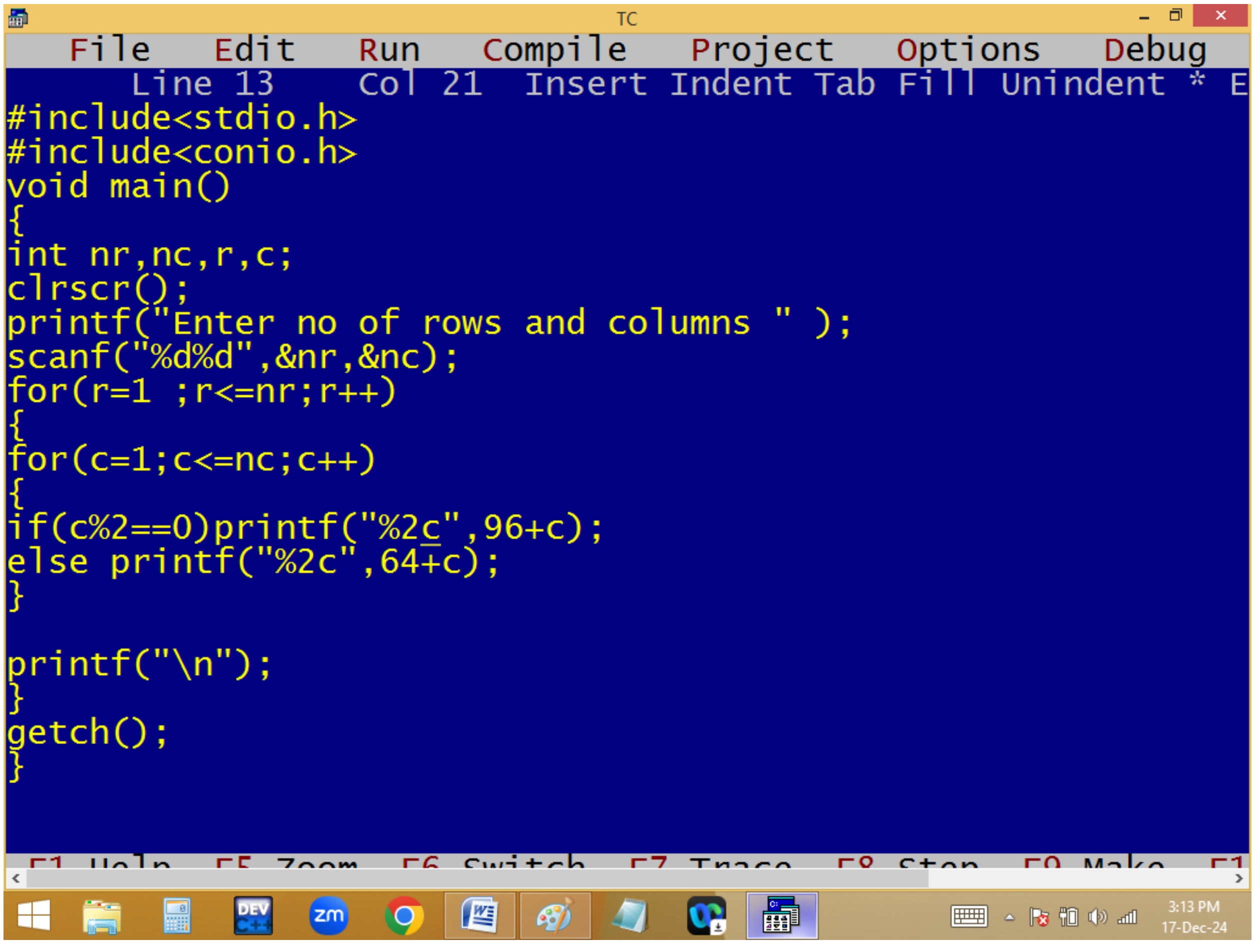



```c
#include<stdio.h>
#include<conio.h>
void main()
{
int nr,nc,r,c;
clrscr();
printf("Enter no of rows and columns " );
scanf("%d%d",&nr,&nc);
for(r=1 ;r<=nr;r++)
{
for(c=1;c<=nc;c++)
{
printf("%3d",r+c-1);
}
printf("\n");
}
getch();
}
```

TC

```
Enter no of rows and columns 5 5
   1   2   3   4   5
   2   3   4   5   6
   3   4   5   6   7
   4   5   6   7   8
   5   6   7   8   9
```

3:07 PM
17-Dec-24


TC
File Edit Run Compile Project Options Debug
Line 13 Col 21 Insert Indent Tab Fill Unindent * E
#include<stdio.h>
#include<conio.h>
void main()
{
int nr,nc,r,c;
clrscr();
printf("Enter no of rows and columns ");
scanf("%d%d",&nr,&nc);
for(r=1 ;r<=nr;r++)
{
for(c=1;c<=nc;c++)
{
if(c%2==0)printf("%2c",96+c);
else printf("%2c",64+c);
}

printf("\n");
}
getch();
}
3:13 PM
17-Dec-24

TC

```
Enter no of rows and columns 10 26
 A b C d E f G h I j K l M n O p Q r S t U v W x Y z
 A b C d E f G h I j K l M n O p Q r S t U v W x Y z
 A b C d E f G h I j K l M n O p Q r S t U v W x Y z
 A b C d E f G h I j K l M n O p Q r S t U v W x Y z
 A b C d E f G h I j K l M n O p Q r S t U v W x Y z
 A b C d E f G h I j K l M n O p Q r S t U v W x Y z
 A b C d E f G h I j K l M n O p Q r S t U v W x Y z
 A b C d E f G h I j K l M n O p Q r S t U v W x Y z
 A b C d E f G h I j K l M n O p Q r S t U v W x Y z
 A b C d E f G h I j K l M n O p Q r S t U v W x Y z
```

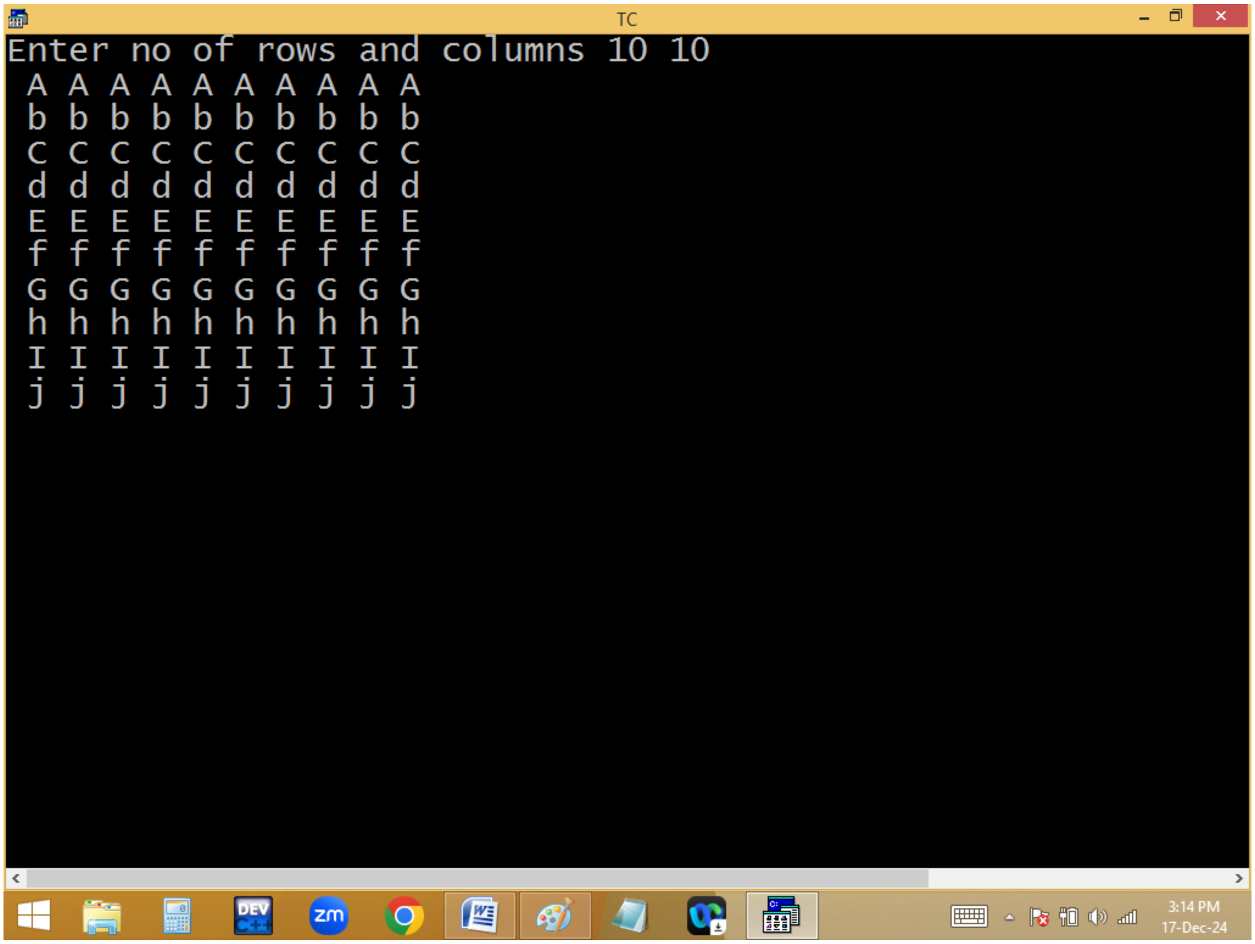
3:13 PM
17-Dec-24

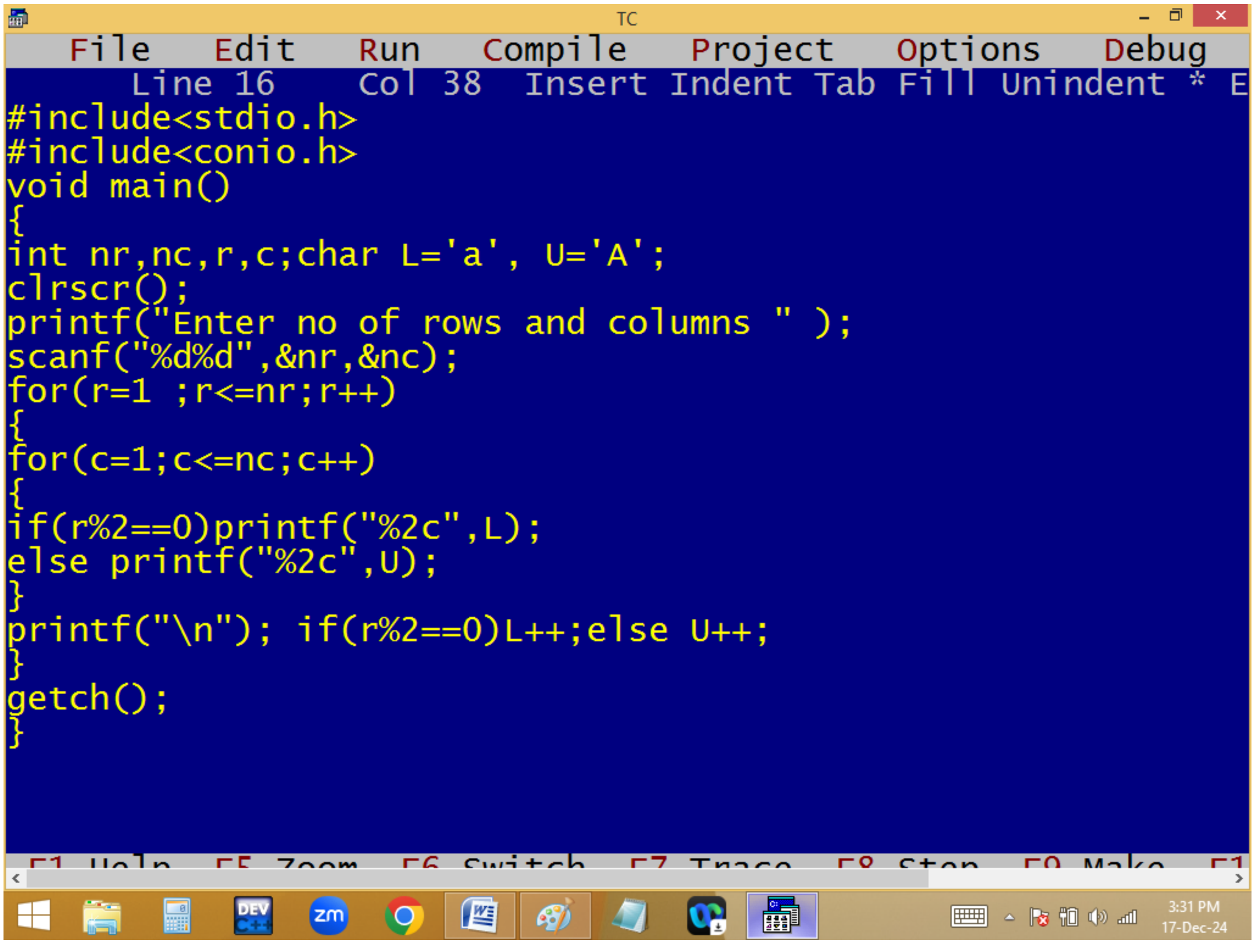
TC

File Edit Run Compile Project Options Debug

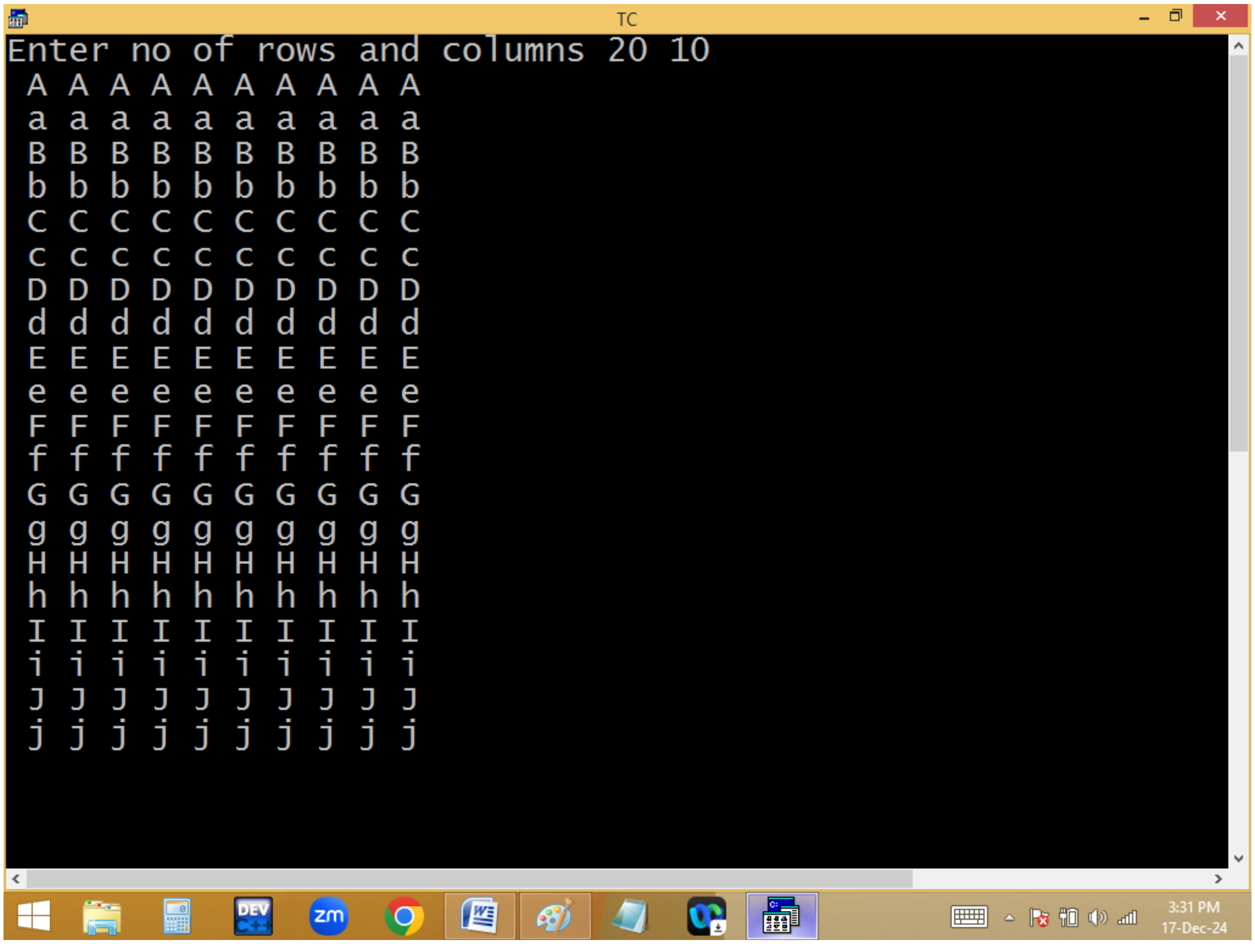
Line 14 Col 23 Insert Indent Tab Fill Unindent * E

```c
#include<stdio.h>
#include<conio.h>
void main()
{
int nr,nc,r,c;
clrscr();
printf("Enter no of rows and columns " );
scanf("%d%d",&nr,&nc);
for(r=1 ;r<=nr;r++)
{
for(c=1;c<=nc;c++)
{
if(r%2==0)printf("%2c",96+r);
else printf("%2c",64+r);
}

printf("\n");
}
getch();
}
```


TC
Enter no of rows and columns 10 10
A A A A A A A A A A
b b b b b b b b b b
C C C C C C C C C C
d d d d d d d d d d
E E E E E E E E E E
f f f f f f f f f f
G G G G G G G G G G
h h h h h h h h h h
I I I I I I I I I I
j j j j j j j j j j
3:14 PM
17-Dec-24

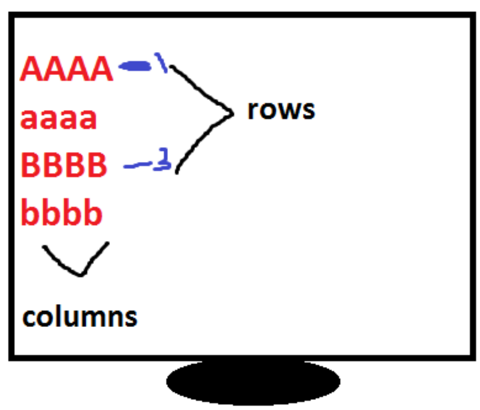



```c
#include<stdio.h>
#include<conio.h>
void main()
{
int nr,nc,r,c;char L='a', U='A';
clrscr();
printf("Enter no of rows and columns " );
scanf("%d%d",&nr,&nc);
for(r=1 ;r<=nr;r++)
{
for(c=1;c<=nc;c++)
{
if(r%2==0)printf("%2c",L);
else printf("%2c",U);
}
printf("\n"); if(r%2==0)L++;else U++;
}
getch();
}
```


```
Enter no of rows and columns 20 10
 A A A A A A A A A A
 a a a a a a a a a a
 B B B B B B B B B B
 b b b b b b b b b b
 C C C C C C C C C C
 c c c c c c c c c c
 D D D D D D D D D D
 d d d d d d d d d d
 E E E E E E E E E E
 e e e e e e e e e e
 F F F F F F F F F F
 f f f f f f f f f f
 G G G G G G G G G G
 g g g g g g g g g g
 H H H H H H H H H H
 h h h h h h h h h h
 I I I I I I I I I I
 i i i i i i i i i i
 J J J J J J J J J J
 j j j j j j j j j j
```

```
char L='a', U='A';
for( r=1; r<=3; r++)
{
for( c=1;c<=4;c++)
{
 if(r%2==0)p(L);
 else  p(U);
}
p("\n"); if(r%2==0)L++;
}          else U++;
```

r c
1 1 2 3 4 5
   * * * *
2 1 2 3 4 5
   * * * *
3 1 2 3 4 5
   * * * *
~~4~~



a

r
1 AAAA
2 aaaa
3
5 bbbb

U
A
B
C

L
a
b

* * * *

* 1 2 *

* 3 4 *

* * * *

1 1 1 1

1 1 2 2

1 3 3 3

4 4 4 4

1 2 2 2

0 1 2 2

0 0 1 2

0 0 0 1